

Uniform effective log Iitaka fibrations for fourfolds

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Abstract

We prove the finite-rational-coefficient case of the effective log Iitaka conjecture in dimension four. For normal projective log canonical complex fourfolds with boundary coefficients in a fixed finite rational set and pseudo-effective rational Cartier adjoint, one uniform degree makes the complete rounded reflexive system nonempty and its section ratios generate the full Iitaka field. We also prove a uniform canonical index bound for projective klt complex fourfolds with rationally trivial canonical divisor.

1 Introduction

Let X be a normal projective klt variety with $K_X \sim_{\mathbb{Q}} 0$. Its canonical index is the least positive integer r for which rK_X is an integral principal divisor. The index problem asks for a bound depending only on the dimension. More generally, for a log Calabi–Yau pair one allows dependence on the boundary coefficients. In dimension three, Jiang obtains a common canonical multiple for all klt Calabi–Yau threefolds, using boundedness modulo flops to handle the noncanonical case [23, Theorem 1.6 and Corollary 1.7]. Xu’s inductive results give the lc threefold index theorem and the klt fourfold theorem with nonzero boundary [41, Theorems 1.13–1.14]. Jiang and Haidong Liu bound surface log pluricanonical representations and, through Fujino’s gluing framework, obtain the index theorem for slc pairs through dimension three and connected non-klt lc fourfold pairs [22, Theorem 1.4 and Corollaries 1.6–1.7]. Birkar also proves an index bound in every fixed dimension for rationally connected klt log Calabi–Yau pairs with coefficients in a fixed rational set satisfying the descending chain condition (DCC) [4, Corollary 1.8]. The distinction between zero and nonzero boundary, and between rationally connected and non-uniruled varieties, is therefore central to the remaining fourfold argument.

Theorem 1.1 (Uniform canonical index). *There is a positive integer N_4 such that $N_4K_X \sim 0$ for every normal projective klt complex fourfold X with $K_X \sim_{\mathbb{Q}} 0$.*

The conclusion concerns integral Weil divisors: it gives a trivialization in one degree for all the fourfolds in the statement. Combined with the preceding non-klt and nonzero-boundary results, it gives the lc fourfold index theorem for any fixed finite rational coefficient set. In applying Xu’s theorem, a finite set is included in the hyperstandard set generated by its complementary coefficients. No assertion for arbitrary real coefficient sets is intended.

Our effective application concerns the map determined by a log canonical divisor. For a rational Cartier divisor D on a normal integral projective variety, fix a divisor representative and put

$$V_m(D) = H^0(X, \mathcal{O}_X(\lfloor mD \rfloor)), \quad m \geq 1.$$

The sheaf here is the rank-one reflexive divisorial sheaf, so this definition does not require mD to be Cartier. The Iitaka field $K(D) \subset \mathbb{C}(X)$ is generated by ratios of nonzero sections in the same $V_m(D)$, over all m . A complete system gives the Iitaka fibration when its ratios generate this entire field. Merely obtaining the correct image dimension would not rule out a finite extension.

Theorem 1.2 (Uniform effective log Iitaka fibrations). *Let $\Phi \subset [0, 1] \cap \mathbb{Q}$ be finite. There is a positive integer $m = m(\Phi)$ such that the following holds. Suppose X is a normal integral projective complex fourfold, $\Delta \geq 0$ is a rational Weil divisor whose nonzero coefficients lie in Φ , (X, Δ) is lc, and $D = K_X + \Delta$ is rational Cartier and pseudo-effective. Then the complete system $|[mD]|$ is nonempty and its section ratios generate $K(D)$ inside $\mathbb{C}(X)$. The same integer works for every choice of canonical divisor on X .*

In Kodaira dimension four the system is birational; in Kodaira dimension zero it is nonempty with image a point. Nonvanishing follows from pseudo-effectivity by the good-model theorem and effective exceptional comparison in OpenAI, *Log abundance in characteristic zero* [34, Theorems 1.1 and 11.1, Lemma 2.2]. This qualitative step supplies a section in some positive degree, not a uniform degree; the uniform $m(\Phi)$ comes from the index and fibration arguments below.

Earlier effective results separate the big case from lower-dimensional Iitaka bases. Hacon–McKernan–Xu give a uniform birational degree for big lc adjoints with coefficients in any fixed DCC set [20, Theorem 1.3]. Viehweg–Zhang treat smooth varieties of Kodaira dimension two. Their degree depends on invariants of the general fibre; for fourfolds the fibre is a surface of Kodaira dimension zero, and boundedness of those invariants makes the degree universal [40, Theorem 0.2 and the discussion preceding Corollary 0.4]. Todorov–Xu treat klt pairs of log Kodaira codimension two in dimensions two, three and four, with coefficients in a rational DCC set [39, Theorem 1.2]. For fourfolds their hypothesis is $\kappa(K_X + \Delta) = 2$. Chen, Han and Jihao Liu formulate the effective log Iitaka conjecture for lc pairs of nonnegative Kodaira dimension with coefficients in an arbitrary DCC subset of $[0, 1]$, and prove it through dimension three [9, Conjecture 1.2 and Corollary 1.5].

The canonical bundle formula relates this question to effective birationality on the base. Birkar–Zhang prove effective Iitaka fibrations for smooth projective varieties of nonnegative Kodaira dimension, with constants depending on the dimension, the least nonvanishing pluricanonical degree on the general fibre, and the middle Betti number of a smooth model of its canonical cover. In fixed dimension and for a fixed DCC coefficient set, their polarized effective birationality theorem gives a uniform degree for a big base adjoint once the nef part has bounded denominator [3, Theorems 1.2–1.3]. Theorem 1.2 gives the fourfold conclusion for finite rational coefficient sets, across all nonnegative Kodaira dimensions and for lc singularities. Its coefficient scope is narrower than that of the DCC conjecture and the earlier big and klt results. The uniform degree uses the canonical-index theorem and the relative denominator theorem below. The passage from pseudo-effectivity to nonvanishing is the separate qualitative input described above.

1.1 The model comparison and the index reductions

The uniform argument needs a model on which the adjoint defines a morphism. The comparison in Proposition 2.2 preserves *every* space $V_m(D)$ under a crepant dlt modification and an adjoint-negative program. Its semiample output uses the abundance-after-nonvanishing theorem of OpenAI, *Lifting sections from the reduced support of an adjoint* [33, Theorem 1.2]. That theorem applies to projective lc rational pairs over $\mathbb{C}$ of dimension at most four: a nef rational Cartier adjoint of nonnegative Kodaira dimension is semiample. In Kodaira dimension zero it is rationally linearly trivial. We state the model comparison before the index reduction because it is needed there as well as in the final effective application.

The singular Beauville–Bogomolov decomposition gives a finite quasi-étale cover by a product of an abelian variety and nonflat Calabi–Yau or symplectic factors. Druel proves the needed dimension-at-most-five result [11, Theorem 1.2]; Höring–Peternell prove the arbitrary-dimensional version [21, Theorem 1.5], using Druel’s integrability results and the holonomy decomposition of Greb–Guenancia–Kebekus [18, Theorem B and Proposition D]. Here quasi-étale means étale outside a subset of codimension at least two. Integral cohomology bounds the abelian characters, and lower-dimensional pluricanonical characters handle products. The remaining single-factor case has rationally connected proper rational images. Its noncanonical quotients are handled by a relative torsion bound; canonical quotients admit a crepant terminal model V with the same canonical index, irregularity zero and countable birational self-map group.

An effective divisor D of Iitaka dimension one, two or three on V would give another bounded-index case. A small rational boundary δD produces a semiample contraction, but the relative index theorem is applied only after proving that the underlying canonical divisors remain crepant. The resulting fibration has boundary zero, so the relative denominator and rationally connected torsion theorems [35, Theorem 1.1 and Proposition 7.1] have constants independent of D and δ . Consequently, in the unbounded-index case every effective divisor of positive Iitaka dimension is big. This implies the precise fibration property used in the scalar estimate: every intermediate equivariant rational fibration of the canonical cover has a geometric generic fibre of general type on a smooth projective resolution. It is a conclusion of the reduction.

1.2 Curves on the canonical cover and its products

Let r be the canonical index of the remaining terminal fourfold V . A rational function trivializing rK_V defines its canonical cyclic cover $\pi : Y \rightarrow V$, a connected degree- r quasi-étale cover with deck group μ_r and trivial canonical class upstairs. The minimality of r is what makes this the primitive cover. Choose an ample rational divisor L on V , scaled so that its volume is bounded above and below by absolute positive constants. Then $(\pi^*L)^4 = rL^4$.

For an ample rational divisor P , two local measurements will be used. The number $\gamma(P; V)$ is the supremum of the order of vanishing of a section in degree k , divided by k , at a very general smooth point; degrees are sufficiently divisible to make the divisor Cartier. Its precise definition is given in Definition 5.3. For an integral curve C through a smooth marked point x , the ratio

$$\frac{P \cdot C}{\text{mult}_x C}$$

compares its degree with its multiplicity at that point. The infimum over such curves is the Seshadri constant $\varepsilon(P; x)$. The scalar estimates bound γ above and ε below by absolute multiples of the fourth root of the volume, under the fibration hypothesis just obtained. Applied upstairs, the lower bound makes every such curve ratio grow at least as $r^{1/4}$.

Now take t copies of Y and divide by the simultaneous deck action:

$$Z_t = Y^t / \mu_{r, \text{diag}}, \quad \theta_t : Z_t \longrightarrow V^t.$$

The polarization P_t is the pullback by θ_t of the sum of the t copies of L . On Z_t , the product estimate gives curves through very general points whose degree divided by marked multiplicity is bounded by a constant times t^2 , independently of r . This estimate concerns the quotient by the diagonal group; it does not assert the same bound on Y^t . The maps are displayed in Figure 1.

The product estimate is proved by reducing a fixed set of data to positive characteristic and comparing ordinary jets with Frobenius jets. The local comparison is that of Mustață–Schwede [32,

$$\begin{array}{ccccc}
Y^t & \xrightarrow{\mu_{r,\text{diag}}} & Z_t & \xrightarrow{\theta_t} & V^t \\
\text{pr}_1 \downarrow & & \downarrow q_1 & & \downarrow \text{pr}_1 \\
Y & \xrightarrow{\pi} & V & \xlongequal{\quad} & V
\end{array}$$

Figure 1: The product polarization is pulled back through θ_t . The geometric generic fibre of q_1 is Y^{t-1} : choosing one lift of the marked point in V removes the simultaneous deck action. This identification allows the growing scalar curve bound upstairs to be tested inside a product whose curve cost is bounded independently of r .

Equation (2.8)]. The canonical filtration described by Sun [37, Theorem 3.7 and Lemma 4.2], in the ideal-theoretic description of Kitadai–Sumihiro [25, Definition 3.1 and Remark 3.2], supplies the truncated symmetric powers and their determinant factors. We prove the slope estimates and the diagonal rank comparison needed for these products. For each variety, cover and product size, the arithmetic model is fixed before the characteristic tends to infinity. The later contradiction fixes a finite list of product sizes before allowing r to grow. These are separate limits; neither requires a uniform arithmetic model for all r .

1.3 From chains of curves to birational maps

The curve estimates concern different spaces. We compare them by forming chains from finitely many covering families of marked curves on Z_t , chosen compatibly with coordinate projections and the group actions. Each movement follows a chosen curve from its marked point to a generic point. The closures of the generically reachable sets are fibres of a rational quotient, called *chain leaves*. The construction uses Campana’s stabilization strategy [8, Theorem 1.1 and its proof sketch]; parameter differentiation in the manner of Ein–Küchle–Lazarsfeld [12, Proposition 2.3] retains a degree bound. In our setting a generic point of an e -dimensional leaf can be reached in at most e movements. If the marked curve ratios are at most B , the leaf degree is at most $(eB)^e$. We prove the projection and finite-quotient compatibilities needed to use these bounds together.

Write $H_t \subset Z_t$ for a generic leaf and $W_t = \theta_t(H_t) \subset V^t$. A positive-dimensional fibre of $H_t \rightarrow V$ would lie, after choosing one lift of the marked base point, in Y^{t-1} . The scalar lower bound there grows with r , contradicting the bounded leaf degree. Thus, for sufficiently large r , the maps to each coordinate are generically finite onto their images and the leaf dimensions lie between one and four. Among a sufficiently long finite list of product sizes, stabilization of dimensions and a comparison of forgetting-map degrees force one map $W_{t+1} \rightarrow W_t$ to have degree one.

This birational forgetting map produces more than a dimension contradiction. The leaf tangent spaces identify positive-dimensional subspaces $A_x \subset T_x V$ at different points. We show that the resulting linear transports depend only on their two endpoints and compose. A vector transported from one fixed point then defines a rational vector field. Degree-one forgetting supplies rational evaluation of its local flow, and inverse flow gives birational self-maps. The resulting uncountable family contradicts the countability of $\text{Bir}(V)$, completing the index argument.

Finally, the model comparison turns the effective theorem into a question on the base of a semiample contraction. For every positive-dimensional base the relative big-base theorem [35, Proposition 6.1] gives the whole base field in a uniform complete floor system, including when the contraction is birational. In Kodaira dimension zero, the three index cases give a nonzero section in one uniform degree. The comparison of all integer degrees brings these systems back to the original

pair. The relative results used in this argument are proved independently of Theorem 1.1. The abundance-after-nonvanishing input from [33] and the uniform and index arguments above do not require logarithmic or orbifold subadditivity. The full pseudo-effective statement of Theorem 1.2, however, first uses [34] to obtain nonvanishing and inherits that companion's logarithmic Iitaka subadditivity input.

1.4 Organization and conventions

Section 2 constructs the semiample model and compares all rounded systems. Section 3 records the exact relative inputs. Section 4 removes product and rationally connected cases and proves the fibration condition needed by the scalar estimate. Sections 5–8 prove the chain, scalar, product and transport statements in that order. Section 9 applies them to the remaining canonical covers. Section 10 derives the effective theorem.

Varieties are over $\mathbb{C}$ unless a geometric generic fibre or an arithmetic reduction is specified. Canonical divisors in a birational or finite comparison are chosen compatibly by rational top forms. We distinguish linear equivalence $\sim$, rational linear equivalence $\sim_{\mathbb{Q}}$, and numerical equivalence $\equiv$. A pair has an effective rational boundary and a $\mathbb{Q}$ -Cartier actual adjoint. Subpairs may have negative boundary coefficients. Log discrepancies are ordinary discrepancies plus one; canonical and terminal singularities for zero boundary are defined by nonnegative and positive ordinary discrepancies on exceptional divisors. We use the usual dlt and slc conventions, including conductor gluing in slc adjunction; see [27].

A contraction is a surjective projective morphism $f : X \rightarrow Z$ of normal varieties with $f_*\mathcal{O}_X = \mathcal{O}_Z$. A rational fibration is understood through a relatively algebraically closed base field inside the total function field. A log Calabi–Yau adjoint is $\mathbb{Q}$ -linearly trivial. A uniform index means a common positive integer giving an integral linearly trivial multiple, not merely a numerical relation. Quasi-étale means finite surjective and étale outside codimension at least two. Rational connectedness on singular spaces refers to a smooth projective model when used in a birational argument.

Divisorial sheaves test rational sections at prime divisors. In particular, $v \neq 0$ belongs to $H^0(X, \mathcal{O}_X(\lfloor mD \rfloor))$ precisely when $\text{Div}(v) + mD \geq 0$. Numerical positivity of a rational divisor always concerns its rational Cartier class. Section-order and volume limits may be taken in sufficiently divisible degrees: powering a section preserves its normalized order. DCC and ACC stand for the descending and ascending chain conditions.

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2 Semiample models and rounded section spaces

We pass from the original pair to a semiample model while preserving every rounded section space. The following comparison keeps every integer degree, even when the corresponding adjoint multiple is not Cartier.

We use the following precise theorem of the companion *Lifting sections from the reduced support of an adjoint* [33, Theorem 1.2].

Theorem 2.1 (Abundance after nonvanishing). *Let (X, Δ) be a projective lc rational pair over $\mathbb{C}$ of dimension at most four, with effective boundary and nef $\mathbb{Q}$ -Cartier adjoint $D = K_X + \Delta$. If $\kappa(X, D) \geq 0$, then D is semiample. If $\kappa(X, D) = 0$, then $D \sim_{\mathbb{Q}} 0$.*

This is the companion's independent after-nonvanishing theorem. Its proof uses the supported-boundary theorem and lower-dimensional abundance; it does not use the separate fourfold nonvanishing companion. The boundary coefficients may be arbitrary rational numbers in $[0, 1]$. The conclusion is qualitative: it supplies neither a uniform index nor a uniform section degree. If a generated Cartier multiple has Iitaka dimension zero, its morphism has image a point and a generating section is nowhere zero, which explains the actual rational linear triviality.

For the original pair we first run an adjoint-negative program and compare all section spaces. For the moving-divisor application in Lemma 4.6, the same procedure applies to a small rational perturbation $(V, \delta D)$. There it supplies only a contraction. Uniform statements enter after the boundary is restored to zero and canonical crepancy is proved separately.

Proposition 2.2. *Let (X, Δ) be a projective lc rational pair of dimension four over $\mathbb{C}$, with $D = K_X + \Delta$ $\mathbb{Q}$ -Cartier and $\kappa(X, D) \geq 0$. There is a projective $\mathbb{Q}$ -factorial dlt pair (X', Δ') with semiample adjoint $D' = K_{X'} + \Delta'$ such that, under the identification of rational function fields and compatible canonical divisors,*

$$H^0(X, \mathcal{O}_X(\lfloor mD \rfloor)) = H^0(X', \mathcal{O}_{X'}(\lfloor mD' \rfloor)) \quad (m \geq 1). \quad (2.1)$$

The coefficients of Δ' belong to those of Δ together with 1. The map is obtained by a crepant dlt modification followed by adjoint-negative birational steps. For a $\mathbb{Q}$ -factorial klt input the latter steps can be taken directly, with klt outputs. The semiample contraction $f : X' \rightarrow Z$ has $\dim Z = \kappa(X, D)$ and

$$D' \sim_{\mathbb{Q}} f^* A$$

for an ample rational Cartier divisor A on the normal projective variety Z .

Proof. We first establish the section comparison used throughout the construction. For a nonzero rational function v and a rational divisor L on a normal variety,

$$v \in H^0(\mathcal{O}(\lfloor mL \rfloor)) \iff \text{Div}(v) + mL \geq 0. \quad (2.2)$$

The equivalence holds because the coefficients of $\text{Div}(v)$ are integers. Effective $\mathbb{Q}$ -Cartier pullback and pushforward give both inclusions for any crepant birational morphism. More generally,

suppose that two models have a common resolution with maps p, q to their input and output, and that their adjoints satisfy

$$p^*D_{\text{in}} = q^*D_{\text{out}} + E, \quad E \geq 0, \quad E \text{ is } q\text{-exceptional}.$$

A section on the output pulls back, remains effective after adding mE , and pushes down to a section on the input. A section on the input pulls back and pushes by q to a section on the output, since $q_*E = 0$. Applying (2.2) proves equality for every m , without assuming that mD is Cartier. Thus the equality survives any sequence of these comparisons.

We now construct the models to which this comparison applies. Take a crepant projective $\mathbb{Q}$ -factorial dlt modification. This is the usual dlt modification theorem; see, for example, [13, Theorem 2.8]. Its boundary consists of the strict transform of Δ and exceptional components of coefficient one.

Run an adjoint-negative minimal model program. We recall the existence and termination inputs needed here. On a fixed negative extremal ray, slightly decreasing the coefficient-one part gives a klt boundary for which that ray is still negative. Klt contraction and flip existence apply [2]. To identify its flip with that of the original adjoint, write the perturbed klt adjoint as D_ϵ and the contraction as g . Relative Picard number one gives a positive rational number a such that $L = D - aD_\epsilon$ is numerically trivial over g . For sufficiently divisible k , both kL and $-kL$ are Cartier, and their differences from D_ϵ are relatively ample. The relative base point free theorem makes both relatively semiample. Their associated maps are constant on each connected fibre of g , so a further multiple of L descends to a Cartier divisor on the normal contraction base. Thus L is a rational pullback. On the small flipped model, the transform of D is consequently a positive rational multiple of the relatively ample transform of D_ϵ plus that pullback. It is relatively ample as required. The original dlt condition is preserved by these negative steps. A fibre-type end is excluded by $\mathbb{Q}$ -linear effectivity of the adjoint, which follows from $\kappa \geq 0$. Indeed, a curve in a general fibre not contained in the support of an effective representative cannot have negative intersection with that representative. Divisorial contractions decrease the Picard number, and the remaining flips terminate by [10, Theorem 1.1], applied with zero nef data to the pseudo-effective rational lc adjoint. The output is therefore nef and $\mathbb{Q}$ -factorial dlt. Starting with a klt pair keeps all these outputs klt.

The initial modification is crepant, and discrepancy comparison for each adjoint-negative step has exactly the effective exceptional form used above. Iteration proves (2.1); in particular the nef output still has nonnegative Kodaira dimension. Theorem 2.1 makes it semiample. A generated Cartier multiple, followed by Stein factorization, gives the stated contraction and ample rational divisor.

We also identify its field with the whole embedded Iitaka field. For sufficiently divisible a , the divisor aD' differs from $f^*(aA)$ by an integral principal divisor. Complete systems of sufficiently high multiples of the ample divisor aA generate $\mathbb{C}(Z)$, so $\mathbb{C}(Z) \subset K(D')$. Conversely, take two nonzero sections in any $V_m(D')$. Raising both to a sufficiently divisible common power puts them in such a pullback degree. Their ratio to that power lies in $\mathbb{C}(Z)$ by $f_*\mathcal{O}_{X'} = \mathcal{O}_Z$. The original ratio is algebraic over $\mathbb{C}(Z)$, which is relatively algebraically closed in $\mathbb{C}(X')$ because f is a contraction. Thus every rounded-degree ratio already lies in $\mathbb{C}(Z)$. Together with (2.1), this proves $K(D) = K(D') = \mathbb{C}(Z)$ inside the common function field. $\square$

Lemma 2.3. *Let D be a rational Cartier divisor on a normal projective variety such that $D \sim_{\mathbb{Q}} f^*A$ for a contraction $f : X \rightarrow Z$ and an ample rational Cartier divisor A . If one nonempty complete system $\llbracket mD \rrbracket$ generates $\mathbb{C}(Z)$ inside $\mathbb{C}(X)$, then every positive integer multiple of m does so as well.*

Proof. The field identification just proved uses only $D \sim_{\mathbb{Q}} f^*A$ and the contraction property, so every rounded-degree ratio lies in $\mathbb{C}(Z)$. Fix a nonzero section s in degree m . For each $k \geq 1$, multiplication by s^{k-1} embeds $V_m(D)$ in $V_{km}(D)$: the pole inequalities add, without a Cartier assumption. Ratios of the resulting sections are unchanged. Hence the degree- km ratios contain the whole field generated in degree m , and cannot generate anything beyond $\mathbb{C}(Z)$. $\square$

3 Relative adjunction inputs

The index reductions use uniform torsion on rationally connected bases; the effective application uses full field generation over a big base. We record the three exact results proved in the companion *Relative denominators and effective systems for log Calabi–Yau fibrations* [35]. Their proofs are independent of the canonical-index theorem proved here and of abundance after nonvanishing.

A generalized pair $(Z, B_Z + M_Z)$ includes a rational b-divisor $\mathbf{M}$ that is the pullback of a nef rational Cartier divisor on a higher projective model. The divisor M_Z is its trace on Z . The expression “ $p\mathbf{M}$ b-Cartier” means that this nef divisor has a Cartier multiple pM_W on one determination W , whose pullbacks give the data on all higher models. Generalized discrepancies are computed after subtracting the nef data on such a model. The following result concerns an exact presentation by rational divisors.

Source: [35, Theorem 1.1].

Theorem 3.1 (Uniform relative denominators). *Fix a finite set $\Phi \subset [0, 1] \cap \mathbb{Q}$ and an integer $1 \leq d \leq 4$. There are positive integers $p_0 \mid p$, with p_0 clearing Φ , and a DCC set $\mathcal{B} \subset [0, 1] \cap \mathbb{Q}$, depending only on d and Φ , with the following property.*

Let (X, B) be a projective lc pair of dimension d , with effective rational boundary having coefficients in Φ . Suppose that $f : X \rightarrow Z$ is a contraction onto a projective base Z , with $\dim Z > 0$, and

$$K_X + B \sim_{\mathbb{Q}} f^*D$$

for a $\mathbb{Q}$ -Cartier divisor D on Z . There exist $\psi \in \mathbb{C}(X)^$ and a $\mathbb{Q}$ -Cartier divisor $D_Z \sim_{\mathbb{Q}} D$ such that*

$$K_X + B + \frac{1}{p_0} \operatorname{Div}(\psi) = f^*D_Z, \quad D_Z = K_Z + B_Z + M_Z. \quad (3.1)$$

Here B_Z is effective with coefficients in $\mathcal{B}$, $(Z, B_Z + M_Z)$ is generalized lc, and $\mathbf{M}$ is b-nef with $p\mathbf{M}$ b-Cartier. If (X, B) is klt, the generalized pair on Z is generalized klt.

The constant p_0 trivializes the geometric generic adjoint, while p controls the moduli divisor on a determination. These are actual divisor statements, not numerical equivalences. A varying coefficient introduced only to construct a contraction will not enter the fixed coefficient set when this theorem is applied with boundary zero.

Source: [35, Proposition 6.1].

Proposition 3.2. *Under the hypotheses of Theorem 3.1, suppose in addition that D is big. There is a positive integer $m = m(d, \Phi)$ such that, for every positive multiple l of m , the complete divisorial system*

$$|[l(K_X + B)]|$$

is nonempty and generates precisely the subfield $\mathbb{C}(Z) \subset \mathbb{C}(X)$. Its rational map is therefore birationally equivalent to f and to the Iitaka fibration of $K_X + B$.

Here the complete floor system generates the entire relatively algebraically closed base field, including its algebraic part. The conclusion holds for every positive multiple of the stated degree. Source: [35, Proposition 7.1].

Proposition 3.3. *Fix an integer $1 \leq d \leq 4$, a DCC set $I \subset [0, 1] \cap \mathbb{Q}$, and a positive integer p . There exists a positive integer $\ell = \ell(d, I, p)$ with the following property. Let Z be a normal projective complex variety of dimension d with a rationally connected smooth projective resolution. Suppose that $(Z, B + M_Z)$ is generalized klt, that $B \geq 0$ has coefficients in I , and that the b -nef rational b -divisor $\mathbf{M}$ has $p\mathbf{M}$ b -Cartier. If*

$$D = K_Z + B + M_Z \sim_{\mathbb{Q}} 0,$$

then ℓD is an integral principal divisor. Consequently, for every integer $a \geq 1$, the divisor $a\ell D$ is principal and

$$h^0(Z, \mathcal{O}_Z(a\ell D)) = 1.$$

The same integer can be chosen for all dimensions in any fixed subset of $\{1, 2, 3, 4\}$.

The last statement will be used directly with zero boundary and zero nef data on a klt variety with rationally connected resolution. It also applies to the generalized base of the first statement. After taking a common multiple with p_0 , its principal divisor pulls back to an integral principal canonical multiple on the total space. This is the only uniform-torsion step in Lemma 4.6.

4 Structural reductions for the canonical index

We reduce the canonical-index theorem to a particular terminal case. First we bound the volume-form characters when the decomposition has several factors or is purely abelian. A remaining single nonflat factor has only rationally connected proper rational images. This bounds the index for noncanonical quotients and for terminal models carrying an effective divisor of intermediate Iitaka dimension. The cases left over have irregularity zero and the fibration property needed for the scalar bounds of Section 6.

All covers and all varieties in this section are connected unless otherwise specified. For a normal variety, $\Omega_X^{[j]}$ denotes the reflexive extension of $\Omega_{X_{\text{reg}}}^j$. The order of $K_X \sim_{\mathbb{Q}} 0$ is the least positive integer r such that $rK_X \sim 0$, with linear equivalence understood for integral Weil divisors.

4.1 Finite covers and canonical characters

We first record the relation between the canonical order and the action on a volume form. Suppose that $q : R \rightarrow X$ is a finite Galois quasi-étale cover of normal projective varieties, with group G , and that $K_R \sim 0$. Choose a generating reflexive top form ω on R . Since a regular function on R is constant, the action on this one-dimensional space has the form

$$g^*\omega = \chi(g)\omega, \quad \chi : G \longrightarrow \mathbb{C}^*.$$

Then

$$mK_X \sim 0 \iff \chi^m = 1. \tag{4.1}$$

Indeed, an invariant rational pluricanonical form on R descends at the function-field level to X . Since q is étale at the generic points of divisors, the descended form has neither zeros nor poles in codimension one when the original form does not. It therefore trivializes the divisorial sheaf $\mathcal{O}_X(mK_X)$. Conversely, a trivializing form downstairs pulls back to a scalar multiple of $\omega^{\otimes m}$, so that

this power is invariant. In particular, bounding the order of χ bounds the exact linear-equivalence order, including the Cartier index.

For any klt X with $K_X \sim_{\mathbb{Q}} 0$, its cyclic index cover is obtained by normalizing the cover of X_{reg} defined by an r -th root of a primitive trivialization of rK_X . Minimality of r , or equivalently the Kummer description of this cyclic extension, makes the cover connected of degree r . It is quasi-étale and has a generating reflexive volume form. The finite discrepancy formula shows that it is klt. Its canonical divisor is Cartier and linearly trivial, so its ordinary discrepancies are integers greater than -1 ; it is consequently canonical. We use the finite discrepancy formula in this form throughout; see [28, Proposition 5.20].

The singular Beauville–Bogomolov decomposition theorem applied to this cover gives a finite quasi-étale cover

$$P = A \times P_1 \times \cdots \times P_s \longrightarrow X, \quad (4.2)$$

where A is an abelian variety, possibly a point, and the positive dimensional P_i are irreducible Calabi–Yau or irreducible holomorphic symplectic varieties in the singular sense [21, Definition 1.4 and Theorem 1.5]. In dimension four the decomposition is already supplied by Druel [11, Theorem 1.2]. The arbitrary-dimensional formulation above combines the algebraic integrability and holonomy methods discussed by Horing–Petersen in the introduction to [21]; in particular they identify the contributions of Druel and of Greb–Guenancia–Kebekus [18, Theorem B and Proposition D]. We use the following precise properties. Each P_i is normal, projective and canonical, with $K_{P_i} \sim 0$ and dimension $d_i \geq 2$. For every connected normal finite quasi-étale cover of P_i , its algebra of reflexive forms is generated by a volume form in the Calabi–Yau case, and by a generically non-degenerate symplectic form in the symplectic case. In particular, there are no reflexive one-forms or reflexive $(d_i - 1)$ -forms on these covers. A one-dimensional factor is included in A .

Proposition 4.1. *There is a uniform positive integer N_{prod} with the following property. If a projective klt fourfold X , with $K_X \sim_{\mathbb{Q}} 0$, admits a cover as in (4.2) having at least two positive dimensional factors, or having only an abelian factor, then $N_{\text{prod}}K_X \sim 0$.*

Proof. **Preserving the product directions.** First suppose that the product has at least two positive dimensional factors. Denote these factors by $B_1, \dots, B_t$, counting the abelian factor, when present, as a single factor. Thus $t \leq 4$, $1 \leq \dim B_i \leq 3$, and at most one B_i is abelian. Take the normal Galois closure $R \rightarrow X$ of the cover, with group G . This is still quasi-étale: over the smooth locus of X the original cover is finite étale by purity, and its Galois closure is finite étale there as well. The map $R \rightarrow P$ is also quasi-étale. In particular, R is canonical and $K_R \sim 0$. Here and below purity is used in its usual form for a finite generically étale map to a regular variety [38, Tag 0BMB].

The product directions pull back, and extend reflexively, to a splitting

$$\mathcal{T}_R = \bigoplus_{i=1}^t \mathcal{E}_i. \quad (4.3)$$

There are no homomorphisms between two distinct summands. To check this, choose a nonabelian factor among the corresponding two factors, say B_i , and fix general smooth points on all the other factors. After shrinking the complementary parameter space, the morphism $R \rightarrow \prod_{j \neq i} B_j$ has normal, pure-dimensional fibres. This follows from normality of the geometric generic fibre in characteristic zero, generic flatness, and openness of geometric normality. Each of its connected components is therefore normal and finite surjective over B_i . The codimension-two locus excluded from the quasi-étale map $R \rightarrow P$ meets a general slice in codimension at least two, so these component covers are quasi-étale.

On the smooth big open of such a component, $\mathcal{E}_i$ restricts to its tangent sheaf and every complementary summand restricts to a trivial bundle. The component has no reflexive one-forms by the defining property of B_i . It has no vector fields either: contraction with its generating volume form identifies its reflexive tangent sheaf with its sheaf of reflexive $(\dim B_i - 1)$ -forms. A homomorphism in either direction between the two summands thus restricts to zero: the resulting forms or vector fields on the big open extend reflexively across its codimension-two complement. These slices sweep a dense open of R , proving the assertion.

Consequently the projections in (4.3) are central idempotents in the finite-dimensional algebra $\text{End}_R(\mathcal{T}_R)$. Its center is a finite-dimensional commutative algebra. Its primitive idempotents determine nonzero direct summands of positive generic rank, whose ranks sum to four; there are at most four of them. The group G permutes these primitive idempotents. The kernel G_0 of that permutation action has index at most $4!$ and fixes every block projection, since each block projection is a sum of primitive central idempotents. It therefore preserves every $\mathcal{E}_i$.

Recovering the factor fields. Let F_i be the relative algebraic closure of $\mathbb{C}(B_i)$ in $\mathbb{C}(R)$, and let $R_i \rightarrow B_i$ be the normalization in this finite field extension. At the generic point, the complementary directions in (4.3) span $\text{Der}_{\mathbb{C}(B_i)} \mathbb{C}(R)$: this follows first on the product and then under the finite separable extension $\mathbb{C}(P) \subset \mathbb{C}(R)$. In characteristic zero the elements annihilated by all these derivations are exactly those algebraic over $\mathbb{C}(B_i)$. Thus G_0 preserves F_i , and acts by birational maps on R_i .

The finite map $R_i \rightarrow B_i$ is quasi-étale. For completeness, the regular extension obtained by adjoining the geometrically integral product of the other factors is linearly disjoint from the finite extension $F_i/\mathbb{C}(B_i)$. Hence the function field of $R_i \times \prod_{j \neq i} B_j$ is an intermediate field between $\mathbb{C}(P)$ and $\mathbb{C}(R)$. This product is normal, so the normalization property gives finite maps

$$R \longrightarrow R_i \times \prod_{j \neq i} B_j \longrightarrow P.$$

Ramification over a prime divisor of B_i would persist after taking this product. Its ramification index would then divide the ramification index at any prolongation to $\mathbb{C}(R)$, contradicting quasi-étaleness of $R \rightarrow P$. Thus R_i is projective and canonical with $K_{R_i} \sim 0$. Choose its generating volume ω_i to be the pullback of a generating volume on B_i .

Bounding the characters in lower dimension. The varieties R_i have dimension at most three. We now use their finite birational actions to bound the characters on their volume lines. Let J_i be the image of G_0 in $\text{Bir}(R_i)$. Resolve the graph of its finite birational action to obtain a smooth projective J_i -variety W_i with a birational morphism to R_i . One can first take the closure of the graph in the product indexed by the group, where the action permutes the factors, and then use functorial resolution [1, Theorem 1.1]. Canonicity implies that the pullback of ω_i is a regular top form on W_i . For every positive integer m ,

$$H^0(W_i, mK_{W_i}) = \mathbb{C} \omega_i^{\otimes m}. \quad (4.4)$$

Indeed, any section on W_i gives a rational pluricanonical form on R_i with no pole at any prime divisor there. It extends reflexively, and $\mathcal{O}_{R_i}(mK_{R_i}) \simeq \mathcal{O}_{R_i}$ has only constant global sections. This also proves that the character in degree m is the m -th power of the character χ_i on the volume line.

Form the finite quotient $q_i : W_i \rightarrow Q_i = W_i/J_i$. Its branch boundary is

$$B_{Q_i} = \sum_D (1 - 1/e_D) D,$$

where e_D is the ramification order along D ; unramified primes have coefficient zero. The quotient ramification formula gives

$$K_{W_i} = q_i^*(K_{Q_i} + B_{Q_i}).$$

The total divisor on the right is $\mathbb{Q}$ -Cartier and the pair (Q_i, B_{Q_i}) is klt. These are the finite quotient and discrepancy properties of a smooth variety with zero boundary [28, Proposition 5.20]; $\mathbb{Q}$ -Cartierness can also be seen by descending a sufficiently divisible local canonical power, or by taking a norm after trivializing near the finite fibre. Since J_i is finite, some power of ω_i is invariant. Galois descent of rational forms, followed by the displayed ramification formula, gives

$$\kappa(Q_i, K_{Q_i} + B_{Q_i}) \geq 0.$$

The coefficients of these boundaries lie in the fixed DCC set $\{1 - 1/e : e \in \mathbb{Z}_{>0}\}$, and $1 \leq \dim Q_i \leq 3$. Effective log Iitaka fibrations in these dimensions [9, Corollary 1.5] therefore give a uniform integer m_* , made common to the three possible dimensions, for which

$$H^0(Q_i, \mathcal{O}_{Q_i}(\lfloor m_*(K_{Q_i} + B_{Q_i}) \rfloor)) \neq 0.$$

A section here pulls back as a rational m_* -canonical form on W_i . The ramification formula and coefficientwise effectiveness show that it has no divisorial poles, so it is a nonzero regular form. It is invariant because it was pulled back from Q_i . Equation (4.4) now implies $\chi_i^{m_*} = 1$.

Combining the factor characters. The wedge of the rational pullbacks of the ω_i to R is the pullback of the product volume on P . It therefore generates K_R , even though R need not itself split as a product. Its character on G_0 is the product of the block characters, and is killed by m_* . A further factor depending only on $[G : G_0] \leq 4!$ kills the character on all of G . For example, one may multiply by the least common multiple of the integers from one to $4!$. Equation (4.1) proves the desired uniform canonical multiple in this case.

The purely abelian case. If $P = A$ is an abelian fourfold, take the same normal Galois closure $R \rightarrow X$. The quasi-étale map $R \rightarrow A$ is étale by purity. A connected finite étale cover of a complex abelian variety is again abelian: analytically it is obtained from a finite-index sublattice of the lattice in its universal cover, and it is projective here. Each element of G acts integrally on $H^4(R, \mathbb{Z})$, a free group of rank $\binom{8}{4} = 70$. Its eigenvalue on the one-dimensional space $H^{4,0}(R)$ is the canonical character. If that eigenvalue has order d , its cyclotomic polynomial divides an integral characteristic polynomial of degree 70, and hence $\varphi(d) \leq 70$. There are only finitely many such d . Their least common multiple kills every value of the character, and (4.1) again gives a uniform trivial multiple of K_X . Taking a common multiple for the two cases proves the proposition. $\square$

4.2 Rational images of a single nonflat factor

Lemma 4.2. *Suppose that a projective klt fourfold X , with $K_X \sim_{\mathbb{Q}} 0$, has a finite quasi-étale cover $P \rightarrow X$ in which P is one four-dimensional irreducible Calabi–Yau or irreducible holomorphic symplectic factor as above. Then the following statements hold.*

- (i) *There is no dominant rational map from X to a positive dimensional non-uniruled smooth projective variety of dimension less than four.*
- (ii) *Every positive dimensional rational image of dimension less than four has rationally connected smooth projective resolution.*
- (iii) *Every smooth projective model of X has irregularity zero.*

Proof. Suppose that $X \dashrightarrow T$ contradicts (i), and set $j = \dim T \in \{1, 2, 3\}$. Non-uniruledness implies that K_T is pseudo-effective by [6]. The minimal-model and abundance theorems in dimension at most three imply $\kappa(T, K_T) \geq 0$. In dimension three, one takes a projective $\mathbb{Q}$ -factorial terminal minimal model and applies abundance for minimal threefolds [26, 24]; canonical sections are unchanged on a resolution. The lower dimensional cases are the corresponding curve and surface statements, or follow by first taking a product with an abelian variety of dimension $3 - j$ and applying the same three-dimensional results.

Choose a nonzero section $\eta \in H^0(T, mK_T)$. Resolve the composite rational map from P to T , obtaining a smooth projective $u : W \rightarrow P$ and a morphism $f : W \rightarrow T$. Pullback by differentials gives a nonzero tensor

$$s = f^* \eta \in H^0(W, (\Omega_W^j)^{\otimes m}). \quad (4.5)$$

It is nonzero because the map is dominant in characteristic zero. At the function-field level it has the form $s = a\beta^{\otimes m}$, where $a \in \mathbb{C}(P)^*$ and β is a nonzero decomposable rational j -form, obtained by pulling back a rational top form on T .

We claim that the divisorial zero locus of s is u -exceptional. Choose an ample Cartier divisor H_P on P , and put $H = u^*H_P$. Since P is canonical with trivial canonical class, K_W is $\mathbb{Q}$ -linearly equivalent to an effective exceptional divisor. Thus it is pseudo-effective and $K_W \cdot H^3 = 0$. Generic semipositivity gives nonnegative degrees for torsion-free quotients of Ω_W^1 with respect to any ample polarization [7, Theorem 2.1]. It gives the same conclusion for torsion-free quotients of $\mathcal{V} = (\Omega_W^j)^{\otimes m}$.

Here is the tensor step explicitly. Use an ample rational perturbation $H + \epsilon H_0$, where H_0 is ample and $\epsilon > 0$. The Harder–Narasimhan factors of Ω_W^1 have nonnegative slopes. On a sufficiently high general complete-intersection curve the filtration restricts to a filtration by semistable bundles [31, Theorem 6.1]; the curve can avoid all the codimension-two loci where the sheaves or quotients fail to be locally free. Tensor products of semistable bundles in characteristic zero are semistable, with the sum of the slopes [36, Theorem 3.18]. Tensor powers consequently have filtrations whose slopes are nonnegative. Exterior powers are quotients of tensor powers, so the same lower slope bound applies to $\mathcal{V}$ and to its torsion-free quotients. Letting ϵ decrease to zero proves the required nonnegative degree against H^3 .

Saturate the rank-one subsheaf generated by s in $\mathcal{V}$. It is a line bundle $\mathcal{O}_W(Z)$, with $Z \geq 0$ the divisorial zero divisor of s , and the quotient is torsion-free. The first Chern class of $\mathcal{V}$ is a positive integral multiple of K_W . It has degree zero against H^3 ; the preceding quotient inequality therefore gives

$$Z \cdot H^3 \leq 0.$$

Every nonexceptional prime divisor has strictly positive intersection with H^3 , by the projection formula and ampleness on P . Since Z is effective, it follows that Z is exceptional, as claimed.

At the generic point of a prime divisor on P , write the rational form β in a regular frame of Ω_P^j , and let $b \in \mathbb{Z}$ be the minimum order of its coefficients. The absence of divisorial zeros of s at that point says

$$\text{ord}(a) + mb = 0.$$

Hence every coefficient of $\text{Div}_P(a)$ is divisible by m . Normalize P in a field component of the extension obtained by adjoining z with $z^m = a$. This cover is finite and quasi-étale: at any divisorial discrete valuation, removing the uniformizer to its m -divisible order reduces the equation to a root of a unit, which is unramified in characteristic zero. The rational form $z\beta$ has no divisorial poles. It is therefore a nonzero reflexive j -form on the cover, and remains decomposable at its generic point.

This contradicts the defining form algebra of P . For a four-dimensional irreducible Calabi–Yau factor there are no such forms in degrees one, two or three on any quasi-étale cover. For a four-dimensional irreducible symplectic factor the only possible degree is two, and every such form is a scalar multiple of the generically nondegenerate symplectic form. Its square is nonzero, whereas the square of a decomposable two-form is zero. This proves (i).

For (ii), resolve the rational image and take its maximal rationally connected fibration. Its smooth projective base is non-uniruled [17, Corollary 1.4]. A positive dimensional such base would be a rational image of X of dimension less than four, contrary to (i). The base must be a point, which is exactly rational connectedness of the resolution.

Finally let W_X be a smooth projective model of X . A regular one-form on W_X pulls back under the dominant generically finite rational map $P \dashrightarrow W_X$ to a reflexive one-form on P . Indeed, the rational map to the proper target is defined at all codimension-one points, and the pulled-back form is regular there; reflexive extension completes it. Pullback is injective at the function-field level in characteristic zero. The absence of one-forms on P forces the original form to vanish. Hodge decomposition and symmetry on the smooth projective W_X then give $h^1(W_X, \mathcal{O}_{W_X}) = h^0(W_X, \Omega_{W_X}^1) = 0$, proving (iii). $\square$

We also need the following numerical fact about exceptional divisors. Its constants depend on the fixed birational morphism only; no uniform birational boundedness is involved.

Lemma 4.3. *Let $u : W \rightarrow X$ be a projective birational morphism from a smooth projective fourfold to a normal projective variety. A pseudo-effective u -exceptional real divisor on W is effective.*

Proof. Fix a very ample Cartier divisor L on X , put $A = u^*L$, and fix an ample Cartier divisor H on W . Because A is big, there are an integer $b > 0$ and an effective Cartier divisor F such that $bA \sim H + F$. If S is a nonexceptional prime divisor not contained in $\text{Supp } F$, then $F \cdot S$ is an effective codimension-two cycle. Both bA and H are nef, and factoring their cubes gives

$$(bA)^3 \cdot S - H^3 \cdot S = F \cdot S \cdot ((bA)^2 + bA H + H^2) \geq 0.$$

There are only finitely many nonexceptional primes in $\text{Supp } F$, and each has positive A^3 -degree. Increasing a constant to handle these primes gives

$$H^3 \cdot S \leq CA^3 \cdot S = CL^3 \cdot u_*S \quad (4.6)$$

for every nonexceptional prime S .

Let $E = \sum_i a_i E_i$ be exceptional and pseudo-effective. Choose effective real divisors D_n whose numerical classes converge to $[E]$. Write $D_n = S_n + T_n$, where S_n has no exceptional components and T_n is supported on the finitely many exceptional prime divisors of u . Projection gives

$$A^3 \cdot S_n = A^3 \cdot D_n \longrightarrow A^3 \cdot E = 0.$$

Equation (4.6) implies $H^3 \cdot S_n \rightarrow 0$. Choose divisor classes $Q_1, \dots, Q_\rho$ forming a basis of $N^1(W)_{\mathbb{R}}$, and choose $c_j > 0$ so that both $c_j H + Q_j$ and $c_j H - Q_j$ are ample. Effectivity gives

$$|Q_j \cdot H^2 \cdot S_n| \leq c_j H^3 \cdot S_n \longrightarrow 0.$$

The Hodge index theorem makes the form $(D, Q) \mapsto D \cdot Q \cdot H^2$ nondegenerate on $N^1(W)_{\mathbb{R}}$. Thus $[S_n] \rightarrow 0$, and consequently $[T_n] \rightarrow [E]$.

The numerical classes of the exceptional prime divisors are linearly independent. In fact, an exceptional numerically trivial linear combination and its negative are both relatively nef. The

negativity lemma forces both to be nonpositive, hence forces the combination to vanish as a divisor [28, Lemma 3.39]. Their positive cone is therefore closed and has unique coordinates in these classes. Since all T_n belong to that cone, so does $[E]$; independence then gives $a_i \geq 0$ for every i . $\square$

Corollary 4.4. *The canonical orders of the noncanonical varieties in Lemma 4.2 are uniformly bounded.*

Proof. Choose a smooth projective resolution $u : W \rightarrow X$ exhibiting a negative ordinary discrepancy. Its canonical divisor satisfies

$$K_W = u^*K_X + E \sim_{\mathbb{Q}} E,$$

where E is exceptional with a negative coefficient. By Lemma 4.3, K_W is not pseudo-effective. The uniruledness criterion of [6] implies that W is uniruled. Its maximal rationally connected quotient consequently has dimension less than four, and its base is non-uniruled [17, Corollary 1.4]. If that base had positive dimension, the induced rational map from X would contradict Lemma 4.2(i). Thus W is rationally connected. Apply Proposition 3.3 directly to $(X, 0)$, with zero nef data. Its hypotheses are precisely klt singularities, $K_X \sim_{\mathbb{Q}} 0$, and a rationally connected smooth resolution, all of which have been checked. Its uniform integer bounds the canonical order. $\square$

4.3 Terminal models and intermediate fibrations

Lemma 4.5. *A normal projective canonical variety X with $K_X \sim_{\mathbb{Q}} 0$ has a projective crepant $\mathbb{Q}$ -factorial terminalization $V \rightarrow X$. The canonical orders of V and X are equal.*

Proof. On a fixed smooth resolution, the discrepancy divisor of X is effective. Any divisor exceptional over that resolution has positive ordinary discrepancy over the smooth resolution, and adding the pullback of an effective discrepancy divisor preserves positivity. Consequently only the finitely many exceptional divisors already on the fixed resolution can have ordinary discrepancy zero over X .

The projective extraction theorem [2, Corollary 1.4.3] extracts this finite list and gives a $\mathbb{Q}$ -factorial variety. One may arrange its klt application as follows. Fix an integer r with rK_X Cartier, and choose an effective ample Cartier divisor H containing all the centers of the selected valuations. Since X is klt, choose $c > 0$ such that (X, cH) is lc. Then, for every divisorial valuation E ,

$$\text{ord}_E(H) \leq A_X(E)/c.$$

Every exceptional log discrepancy greater than one is at least $1 + 1/r$. Choosing a positive rational δ with $\delta/c < 1/(r + 1)$ keeps all those discrepancies greater than one, while the selected log discrepancies equal to one decrease strictly below one. Take δ still smaller if necessary so that $(X, \delta H)$ is klt, and apply the extraction theorem to this pair and exactly the selected list. The list can be empty, in which case the theorem gives a small $\mathbb{Q}$ -factorialization.

Restore boundary zero. The extracted divisors have ordinary discrepancy zero for X , so the resulting morphism is crepant. All divisors still exceptional over the extracted variety have positive ordinary discrepancy; hence the variety is terminal. Finally, compatible canonical divisors satisfy $K_V = u^*K_X$. A principal multiple downstairs pulls back to the same principal multiple upstairs, and a principal multiple upstairs pushes forward to the same one downstairs. This proves equality of their exact orders. $\square$

Lemma 4.6. *There is a uniform positive integer N_{mov} with the following property. Let V be a projective $\mathbb{Q}$ -factorial terminal fourfold with $K_V \sim_{\mathbb{Q}} 0$, and suppose that every positive dimensional rational image of dimension less than four has rationally connected smooth projective resolution. If V has an effective Weil divisor D with $1 \leq \kappa(V, D) \leq 3$, then $N_{\text{mov}}K_V \sim 0$.*

Proof. By $\mathbb{Q}$ -factoriality, D is $\mathbb{Q}$ -Cartier. Choose a small positive rational number δ for which $(V, \delta D)$ is klt. Its adjoint has Iitaka dimension $\kappa(V, D)$, since $K_V \sim_{\mathbb{Q}} 0$. Apply Proposition 2.2, starting its birational program at V , to obtain a $\mathbb{Q}$ -factorial klt model V' and its semiample contraction

$$f : V' \longrightarrow Z, \quad 1 \leq \dim Z = \kappa(V, D) \leq 3.$$

We check the canonical divisor separately from the perturbed adjoint. Choose compatible canonical representatives and a rational function h with $rK_V = \text{Div}(h)$. Each birational contraction in the program pushes this same relation to the target, and each flip carries it to the new model by its codimension-one identification. All these models are $\mathbb{Q}$ -factorial. On a common resolution their canonical pullbacks are therefore the same divisor $r^{-1} \text{Div}(h)$. In particular, $K_{V'} \sim_{\mathbb{Q}} 0$, the zero-boundary varieties are crepant, and their canonical orders are equal. They remain klt; alternatively, this follows from crepant equality starting at the terminal V .

Now use Theorem 3.1 on the contraction $f : (V', 0) \rightarrow Z$, with zero boundary and base class zero. It gives uniform integers $p_0 \mid p$, a rational function ψ , and generalized klt data satisfying

$$K_{V'} + \frac{1}{p_0} \text{Div}(\psi) = f^* D_Z, \quad D_Z = K_Z + B_Z + M_Z \sim_{\mathbb{Q}} 0, \quad (4.7)$$

where $B_Z \geq 0$ has coefficients in a fixed DCC set and the nef $\mathbb{Q}$ -divisor has Cartier denominator dividing p . The base Z is a rational image of V , so its smooth projective resolution is rationally connected. Proposition 3.3 gives a uniform l , which we enlarge to a multiple of p_0 , such that $lD_Z \sim 0$. Multiplying (4.7) by l then gives $lK_{V'} \sim 0$, hence $lK_V \sim 0$.

The coefficient δ and the divisor D were used only to produce a semiample model. The two uniform statements were applied afterward with boundary exactly zero, in one of the three fixed base dimensions. Their integers thus depend on neither D nor δ . A common multiple for these dimensions is N_{mov} . $\square$

Lemma 4.7. *Let V be a projective $\mathbb{Q}$ -factorial terminal fourfold with $K_V \sim_{\mathbb{Q}} 0$. Suppose that every effective Weil divisor on V of positive Iitaka dimension is big. Let $\pi : Y \rightarrow V$ be its cyclic index cover, with deck group G . Then every G -equivariant rational fibration of Y with positive dimensional base and fibre has a geometric generic fibre of general type on a smooth resolution. The same conclusion holds for every such rational fibration of V .*

Proof. The variety Y is terminal. In fact, an exceptional divisorial valuation over Y restricts to an exceptional valuation over V , and the finite discrepancy formula multiplies its log discrepancy by its positive ramification index. The log discrepancy remains strictly greater than one. Also $K_Y \sim 0$.

Consider a G -equivariant rational fibration $Y \dashrightarrow T$. Its base field $\mathbb{C}(T)$ is relatively algebraically closed in $\mathbb{C}(Y)$. The invariant field $\mathbb{C}(T)^G$ lies in $\mathbb{C}(Y)^G = \mathbb{C}(V)$, has the same positive transcendence degree as $\mathbb{C}(T)$, and is relatively algebraically closed in $\mathbb{C}(V)$. For the last assertion, an element of $\mathbb{C}(V)$ algebraic over $\mathbb{C}(T)^G$ is algebraic over $\mathbb{C}(T)$, hence belongs to $\mathbb{C}(T)$, and then is invariant. A projective model of the invariant field therefore gives a descended dominant rational map $V \dashrightarrow Z$, where $0 < \dim Z < 4$.

Fix smooth projective resolutions of the maps and the bases. Thus we may use a smooth projective birational morphism $u : U \rightarrow Y$, a morphism $f : U \rightarrow T'$ to a smooth model of the unquotiented base, and a morphism $g : U \rightarrow Z'$ to a projective model of the quotient base. The map g is constant on the generic fibre of f . Take a general hyperplane divisor H for a very ample embedding of Z' . On the domain of $V \dashrightarrow Z'$, pull it back and take its closure to obtain an effective Weil divisor D on V . The domain contains every codimension-one point, because V is normal and Z' is proper. The hyperplane pulls form a moving system of linearly equivalent divisors: the

rational-function ratios on the domain extend as the same principal-divisor relations on V . Some ratio is nonconstant, since $\dim Z' > 0$. Thus $\kappa(V, D) \geq 1$, and D is big by hypothesis.

Write $q = \pi \circ u$. We have

$$q^*D = g^*H + E, \quad (4.8)$$

where E is an effective u -exceptional rational divisor. The pullback exists because V is $\mathbb{Q}$ -factorial. At every nonexceptional generic divisorial point of U , the two terms q^*D and g^*H agree: the finite cover takes that point to a codimension-one point of V , where the original rational map is defined. Thus their difference is exceptional. After the resolution has been fixed, choose H general enough that it contains none of the generic images under g of its finitely many exceptional prime divisors. The coefficient of g^*H at each such prime is then zero, while q^*D is effective. This proves the claimed effectivity of E .

Let F be the smooth geometric generic fibre of f . The divisor q^*D is big on U , and its restriction to F is big as well. Indeed, write a big rational class as the sum of an ample rational class and an effective rational class. The integral geometric generic fibre is not contained in the support of the effective term, and the ample term restricts amply. The class g^*H restricts trivially to F , so (4.8) implies that $E|_F$ is big.

Terminality and $K_Y \sim 0$ give

$$K_U \sim_{\mathbb{Q}} \sum_i a_i E_i, \quad a_i > 0,$$

where the sum runs over all u -exceptional prime divisors. Since E is effective and supported on this finite set, there is a positive rational number c such that $\sum_i a_i E_i - cE \geq 0$. Restriction to the geometric generic fibre is effective. Smooth-fibre adjunction identifies $K_U|_F$ with K_F , and therefore K_F is the sum, up to $\mathbb{Q}$ -linear equivalence, of the big divisor $cE|_F$ and an effective divisor. It is big, which proves that F is of general type.

For a rational fibration of V , use the identical argument with $Y = V$, $\pi = \text{id}$, and the trivial group. Terminality of V provides the same positive discrepancy comparison. $\square$

We have now removed the product, noncanonical and intermediate-divisor cases by uniform index bounds. In a hypothetical sequence of unbounded indices, its remaining terminal models V therefore have irregularity zero and no effective nonbig divisor of positive Iitaka dimension. Lemma 4.7 gives the requisite general-type fibres for both V and its canonical cover Y . The next three sections establish chain degree bounds and the scalar and product curve estimates for these varieties; Section 8 will turn their discrepancy into birational self-maps, and Section 9 will finish the contradiction.

5 Generic chains and orders of sections

The canonical-index argument will produce curves of uniformly bounded degree divided by their marked multiplicity on finite product covers. We need to replace those curves by the fibres of a rational map and retain quantitative degree bounds under projection and finite quotient. This section carries out that step. Its order propagation uses the parameter-differentiation method of Ein–Küchle–Lazarsfeld [12, Proposition 2.3], applied here along generic chains. For the quotient we use the stabilization strategy in Campana’s chain-equivalence construction [8, Theorem 1.1 and its proof sketch], with the generic-point formulation and quantitative estimates required here. We prove the projection and finite-quotient compatibilities as well.

We work in characteristic zero and first specify the generic-point conventions used in the arguments. All finite collections of varieties, line bundles, morphisms, and sections over $\mathbb{C}$ descend to countable subfields. After enlarging such a field and taking its algebraic closure, we may take successive geometric generic points inside a sufficiently large algebraically closed extension. A point chosen generically at a particular stage is generic over the field of all the data already chosen. We always specify a smaller field when taking the closure of that point.

Conditions on jet-kernel dimensions involve only countably many pairs of integers: a degree and an order. For each pair, generic base change and semicontinuity on smooth opens give the corresponding generic condition. Thus their simultaneous validity at a geometric generic point is consistent with the usual very-general-point convention over $\mathbb{C}$. Countably many exceptional closed sets can be included in the initial field of definition. Conversely, finite data at a geometric generic point, including a section or a marked curve, can be realized over $\mathbb{C}$ while preserving its genericity over that field.

We also use the following spreading convention. If new finite-type data are chosen on a geometric generic member of a family, they are defined over a field finitely generated over its function field. Spreading that field gives a *dominant* extension of the parameter space, and the new geometric generic fibre realizes the chosen data. Resolutions and graphs can be spread in this fashion. Section spaces and their jet kernels commute with algebraically closed field extension. No assertion about constancy of all-degree section invariants on special fibres is intended.

5.1 The rational quotient generated by marked families

Fix an integral projective variety V . A *marked covering family* on V consists of an integral parameter space S , a dominant marked-point map $u : S \rightarrow V$, and a family of subvarieties $C_b \subset V$ containing $u(b)$. After restricting to a dense open of S , we require its geometric generic member to be integral of positive dimension. We use the reduced incidence in $S \times V$; its endpoint evaluation dominates V because it contains the graph of u . All generic choices below are made on these opens.

Starting from a generic point a of V , a *generic movement* chooses one of the marked families, a generic point b of a geometric component of $u^{-1}(a)$ over the field of the existing path, and then a generic endpoint on C_b over that field together with b .

Lemma 5.1 (Generic chains). *Let a nonempty finite list of marked covering families be given on an integral projective variety V in characteristic zero. There is a dominant rational map*

$$\xi : V \dashrightarrow M$$

with geometrically integral generic fibre, whose fibre closures $H_p \ni p$ have a constant positive dimension e for general p . Starting at a generic p , a path of at most e generic movements reaches a point generic in H_p over the algebraic closure of the field of p . Every generic movement preserves the value of ξ . In particular, a generic family member C_b is contained in $H_{u(b)}$.

The assignment is equivariant under any finite automorphism group preserving the family list.

Proof. Choose an algebraically closed field of definition k_* and a point p generic in V over k_* . For a tuple a of points write $K_a = \overline{k_*(a)}$ inside one sufficiently large algebraically closed extension. Fields for successive choices include the entire preceding path.

Every point encountered in a generic path is individually generic in its ambient space over k_* . Indeed, over a generic mark of V , all components of the geometric generic fibre of a dominant map $S \rightarrow V$ have dimension $\dim S - \dim V$. A parameter generic in such a component is therefore generic in S . The endpoint is then generic in the incidence with that parameter, and hence in V .

We will compare paths after enlarging K_p . The closure of a finite path tuple over the algebraically closed field K_p is geometrically integral. A generic point of its base change realizes the same path after an algebraically closed field extension, with the same projected loci. For a fixed sequence of families, a prefix of length j has total transcendence degree

$$\sum_{i=1}^j (\dim S_i - \dim V + \dim C_{b_i})$$

over its initial-point field: these are precisely the increments from the parameter and endpoint choices. Any tuple satisfying those incidences has at most this transcendence degree. A path generic after an algebraically closed field extension has the displayed degree there, and cannot have a smaller degree before the extension. Thus all prefix degrees agree with their upper bounds, proving the converse comparison as well. This observation also applies when a path is begun at a later endpoint and earlier path data are subsequently adjoined to its ground field.

Let d_j be the largest dimension of the closure *over* K_p of an endpoint after j generic movements; set $d_0 = 0$. These loci can be obtained from components of the finite-step path spaces and their projections. Appending a step makes the new endpoint locus contain the old one. To see this, the new locus contains, after extension to the field of the preceding path and the new parameter, the whole integral member of that last step. It therefore contains the previous endpoint, and hence its closure over K_p . Consequently d_j is nondecreasing, $d_1 > 0$, and the first equality has the form

$$d_h = d_{h+1} = e, \quad 1 \leq h \leq e \leq \dim V.$$

Choose an endpoint locus H_p of dimension e at length h and a corresponding generic endpoint y .

Every possible generic next movement, made over the preceding path, has its member and endpoint in H_p : its endpoint locus contains H_p , and the equality of dimensions forces equality. By the field-extension observation this is also true for all generic choices over $K_{p,y}$, without retaining the rest of the path. The new endpoint is again generic in H_p over K_p . For each family and each geometric component of its parameter fibre over the generic point of H_p , choose the new parameter independently of the previous path. The stabilized dimension calculation says that the endpoint image of this generic incidence is contained in H_p . This is an identity on the incidence over H_p , so it persists after each subsequent independent generic base change. Induction therefore keeps all later paths in H_p , with endpoints generic there over K_p . We use fresh generic realizations in this argument; a previously chosen endpoint need not remain generic after arbitrary old data are adjoined.

Now form endpoint loci of paths intrinsically over K_y , and realize their generic choices also over $K_{p,y}$. They are contained in H_p . Because y is generic in V over k_* , their maximum dimension at length h is the same number e . Every maximizing locus therefore becomes H_p after extension. This proves uniqueness of the maximizing locus at y . Transporting the statement between generic points of V proves uniqueness at a generic starting point and shows

$$H_y = H_p \quad \text{when } y \text{ is generic in } H_p \text{ over } K_p. \quad (5.1)$$

Uniqueness gives descent to $k_*(p)$. More explicitly, choose a projective embedding over k_* and take the equation subspaces of the saturated homogeneous ideal of H_p through a degree generating that ideal. Every algebraic conjugate is again the unique maximizing locus. Its Grassmann coordinates are therefore invariant, and belong to $k_*(p)$ in characteristic zero. They define a rational map ξ to the closure of their image, naming H_p . We may replace the image by its normalization. Equality of names means equality of subvarieties.

For a next step from the point y just considered, both endpoints are generic in H_p over K_p , so (5.1) gives equality of their names. Realizing each choice generically after field extension shows that this equality holds at a generic parameter and endpoint of *each* family. It is therefore a rational identity on its incidence. It follows that every generic movement from a generic point preserves ξ , including movements that are not part of the original maximizing path.

For completeness, fix a generic name $m = \xi(p)$ and work over K_m . A generic point of any component of the geometric fibre of ξ is generic in V over k_* . It belongs to the subvariety bearing its own name, so each such component is contained in the integral subvariety named m . Conversely, a generic point of that subvariety, chosen also generically over K_p , has name m by (5.1). Thus the closures of these fibres equal that integral subvariety. Characteristic zero gives geometric reducedness as well. The graph closure supplies a rational family with geometrically integral generic fibre; geometric integrality and constant dimension spread to an open of the base, by generic flatness and the constructibility of geometric fibre properties [19, §§9,12]. This proves the fibre assertions. Finally, a finite automorphism preserving the family list takes a maximizing locus to a maximizing locus. Uniqueness proves equivariance. The induced rational action on the name space can, if desired, be made regular by taking the closure of its graph in the product of the finitely many translates. This leaves the generic leaves unchanged. $\square$

5.2 Projection and finite-quotient compatibility

Corollary 5.2. *Suppose a dominant rational map between two ambient varieties sends generic movements of an upper family list to generic movements of a lower family list, or to stationary steps. Then the image of a generic upper leaf is contained in the corresponding lower leaf.*

If a finite group preserves a family list on V and $\theta : V \rightarrow V'$ is its quotient, the subvarieties $\theta(H_p)$ are the closures of the geometrically integral generic fibres of a rational fibration on V' .

Proof. The lower rational name is unchanged along every projected step. Following a path that fills the upper leaf proves the first assertion. It is enough here that equality of the lower names holds at the generic point of each projected incidence: that rational identity persists when the step is realized over the additional upper path data.

For the second assertion, let p' be generic in V' and lift it to $p \in V$. All lifts are conjugate, and equivariance gives the same image

$$W_{p'} = \theta(H_p)$$

for every lift. Because p is algebraic over p' , their algebraic closures in the fixed overfield agree: $K_p = K_{p'}$. The image equation subspaces descend to $k_*(p')$ by conjugation invariance, just as in the proof of Lemma 5.1. They give a rational name for $W_{p'}$. A point generic in $W_{p'}$ over $K_{p'}$ lifts to a point generic in H_p over K_p , and hence has the same name as p' . Conversely every ambient generic point belongs to the subvariety with its own name. The preceding fibre-of-the-name argument proves geometric integrality and identifies its fibre closure with $W_{p'}$. $\square$

5.3 From marked-member bounds to leaf degree

The quotient and its compatibilities depend only on the marked families. We now put a polarization on the ambient variety. The estimate below bounds the degree of a leaf by propagating section orders along a path that fills it. We first define the section-order invariant used in this argument and later in the scalar estimates.

Definition 5.3. Let V be an integral projective variety of positive dimension, and let P be a big, nef, semiample rational Cartier divisor on V . For a smooth projective birational resolution $\pi : \tilde{V} \rightarrow V$, set

$$\gamma(P; V) = \sup_{\substack{k > 0 \text{ sufficiently divisible} \\ 0 \neq s \in H^0(\tilde{V}, k\pi^*P)}} \frac{\text{ord}_x(s)}{k}, \quad \rho(P; V) = (P^{\dim V})^{1/\dim V},$$

where x is very general in the smooth isomorphism locus, identified with its lift to $\tilde{V}$. If a specified smooth point x is used instead, write $\gamma(P; V, x)$, with the resolution chosen to be an isomorphism near x . For a subvariety A , the notation $\gamma(P; A)$ means the same invariant for the restriction of P , pulled back to a resolution of A .

These definitions do not depend on the chosen smooth resolution: proper birational push-forward of the structure sheaf between normal varieties and the projection formula preserve the pulled-back section spaces. Common resolutions then compare any two choices. They also allow any additional fixed divisibility requirement on k , since taking a power of a section preserves its normalized order. The invariant is finite: on a smooth model, restrict a given nonzero section to a general complete-intersection curve through the mark, chosen not to lie in its zero divisor. Its order is at most the degree of the restricted line bundle, which is a fixed constant times k . The invariant is positively homogeneous under rational scaling of P and is unchanged by rational linear equivalence at a very general mark.

We will repeatedly use

$$\gamma(P; V) \geq \rho(P; V). \quad (5.2)$$

Indeed, put $n = \dim V$ and choose $q > 0$ such that $q\pi^*P$ is Cartier and basepoint-free on $\tilde{V}$. Its morphism f to its projective image is generically finite by bigness. The Hilbert polynomial of $f_*\mathcal{O}_{\tilde{V}}$, twisted by the hyperplane bundle of the image, gives

$$h^0(\tilde{V}, qt\pi^*P) = \frac{P^n}{n!}(qt)^n + O(t^{n-1}).$$

The leading coefficient is the generic rank of $f_*\mathcal{O}_{\tilde{V}}$ times the degree of the image, divided by $n!$; it equals $(q\pi^*P)^n/n!$ by the projection formula. Vanishing to order at least l at a smooth point imposes at most $\binom{n+l-1}{n}$ conditions. Taking l/k below, and then arbitrarily close to, $(P^n)^{1/n}$ proves (5.2).

Lemma 5.4 (Orders and degrees of chain leaves). *Let the marked families and their rational quotient be as in Lemma 5.1, and let e be the dimension of a generic leaf. Suppose that P is a big, nef, semiample rational Cartier divisor on V . Assume that, for every family, its geometric generic marked member satisfies one of the following bounds, with a common number $B \geq 0$:*

- (i) *the mark is smooth on C_b and $\gamma(P; C_b, u(b)) \leq B$;*
- (ii) *C_b is a curve and $P \cdot C_b / \text{mult}_{u(b)} C_b \leq B$.*

Then on the geometric generic leaf,

$$\gamma(P; H_p) \leq eB, \quad P^e \cdot H_p \leq (eB)^e. \quad (5.3)$$

Proof. Choose an algebraically closed field of definition k_* for the data. Let m be a generic point of the quotient base and put $K_m = \overline{k_*(m)}$ inside the fixed algebraically closed overfield. The restriction of P to the geometric generic leaf over K_m is big: the leaf sweeps V , so it meets the generically

finite locus of the morphism defined by a basepoint-free multiple of P . Take a smooth projective resolution J of this leaf over K_m . The generic marked families and paths lift to J wherever the resolution is an isomorphism. A point generic in the leaf over K_m is generic in V over k_* . For a generic mark p on this leaf, put $K_p = \overline{k_*(p)}$. Its name m belongs to $k_*(p)$. Lemma 5.1 supplies a path of length $h \leq e$ reaching a point generic in the whole leaf over K_p . Every intermediate mark and endpoint is generic in J over K_m : it is generic in V and has name m , and consequently has transcendence degree e over K_m . In particular all their evaluation maps dominate J .

Suppose that, in a divisible degree k , a nonzero section of kP has order at least an integer $m_0 > hkB$ at the variable generic mark p of J . In a fixed K_m -basis of $H^0(J, kP)$, the kernel of the generic jet-evaluation matrix lets us choose such a section $\sigma(p)$ with coefficients rational in p . Follow a path of length h that fills the leaf generically over K_p .

Assume inductively that the same section has multiplicity at least m_j at the current mark. Let T be the closure over K_m of the path tuple through the parameter b chosen for the next step; the original point p varies in this closure. Let $I \subset T \times J$ be the incidence obtained by adjoining the new endpoint. Both the graph of the current mark and I dominate J . Over the generic point of T , the generic fibre of I is the chosen member, lifted to J .

Differentiate in parameter directions, keeping the variable on J fixed. Concretely, use derivations of $K_m(T)$ over K_m on the coefficients of σ in the fixed section basis. A line-bundle frame from J is regular and nonvanishing at the generic point of J , so it is unaffected by these derivations. At the graph of the moving mark, the multiplicity assumption is membership in the m_j th power of the graph ideal, after localizing over the generic parameter. Parameter differentiation drops its powers by at most one. This can also be tested after field extension to make the mark rational, using faithful flatness.

Put

$$r_j = m_j - \lfloor kB \rfloor.$$

Every iterated derivative of length less than r_j still has order strictly greater than kB at the moving mark. Each such derivative vanishes identically on the chosen member. In case (i), a nonzero restriction, pulled back to its smooth resolution, would violate the assumed order bound at the smooth mark. The resolutions used in this comparison agree near the mark and can otherwise be compared on a common resolution.

In case (ii), a section with order at least l at the ambient mark restricts to an element of the l th power of the maximal ideal of the local curve ring A . On its normalized branches its order is at least $l v_i(\mathfrak{m}_A)$. The identity

$$\text{mult } A = \sum_i v_i(\mathfrak{m}_A)$$

then shows that a nonzero restriction has degree at least $l \text{ mult } A$. One way to verify this identity is to use the finite semilocal normalization $\bar{A}$. Its quotient by $\mathfrak{m}_A^t \bar{A}$ has length $t \sum_i v_i(\mathfrak{m}_A)$ over the algebraically closed residue field. The conductor gives a fixed c such that

$$\mathfrak{m}_A^{t+c} \bar{A} \subset \mathfrak{m}_A^t A \subset \mathfrak{m}_A^t \bar{A}.$$

Comparing leading terms in the Hilbert–Samuel lengths gives the identity. Here $l > kB$, whereas the degree of a nonzero restricted section is $kP \cdot C_b \leq kB \text{ mult } A$, a contradiction.

Thus all parameter derivatives of length less than r_j vanish at the generic point of I . We explain why this forces order at least r_j along that incidence. Let

$$R = \mathcal{O}_{T \times J, \eta_I}, \quad F = K_m(J).$$

Since I dominates both factors, this ring localizes over their generic points. It is regular, using a smooth open of T . The conormal map

$$\mathfrak{m}_R/\mathfrak{m}_R^2 \longrightarrow \Omega_{R/F} \otimes \kappa(I)$$

is injective. Indeed the residue extension is separable in characteristic zero; in the cotangent sequence the dimensions of the ambient differentials and residue differentials differ by $\dim R$, which is the conormal dimension. Equivalently, this is the generic smoothness computation for the incidence over J . Parameter derivations span the dual of this conormal after reduction, since R is a localization of $K_m(T) \otimes_{K_m} F$. Regularity identifies the associated graded ring with the symmetric algebra of the conormal. A nonzero initial form of degree $a < r_j$ is detected by a length- a iterated parameter derivative, by the symmetric-power pairing and the invertibility of $a!$. The derivative vanishings therefore imply that our section, in the chosen frame, lies in $\mathfrak{m}_R^{r_j}$.

Pass to the generic parameter field and then extend it so that the independently generic endpoint is rational. The map from the ambient product to this fibre induces a local homomorphism from R to the local ring of J at that endpoint. A denominator outside the incidence prime has nonzero value at its generic point and stays a unit after this extension. The incidence ideal maps into the endpoint maximal ideal; its r_j th power therefore maps into that maximal ideal's r_j th power. The same section consequently has multiplicity at least

$$m_{j+1} = r_j = m_j - \lfloor kB \rfloor$$

at the next endpoint. After h steps it still has positive multiplicity, since $m_0 > hkB$. But the endpoint is generic in J over K_p , while $\sigma(p)$ is a nonzero section defined over K_p . This is impossible. Hence $\gamma(P; H_p) \leq hB \leq eB$. Applying (5.2) on the leaf gives the second inequality of (5.3). $\square$

6 Scalar jet bounds

The next result compares section orders and curve degrees with the volume of an ample rational divisor. Its hypothesis on equivariant fibrations is essential to both estimates. In the index argument, Lemma 4.7 establishes this hypothesis on the varieties to which we apply the result.

We use the normalized order invariant of Definition 5.3. The following estimates apply to rational polarizations of arbitrary denominator. All orders and section spaces are taken in sufficiently divisible degrees; taking powers shows that imposing any further fixed divisibility does not change the normalized supremum.

Theorem 6.1 (Scalar bounds). *There are constants $C, c > 0$ with the following property. Let V be a normal projective canonical fourfold with $K_V \sim_{\mathbb{Q}} 0$, and let a finite group G act on V . Suppose that every G -equivariant rational fibration with geometrically integral generic fibre and with both base and fibre of positive dimension has geometric generic fibre of general type on a smooth projective resolution. It is enough to impose this condition on the rational quotients supplied by Lemma 5.1. If P is an ample rational Cartier divisor satisfying $g^*P \sim_{\mathbb{Q}} P$ for every $g \in G$, then*

$$\gamma(P; V) \leq C(P^4)^{1/4}, \quad \varepsilon(P; x) \geq c(P^4)^{1/4} \tag{6.1}$$

at a very general smooth point $x \in V$, where

$$\varepsilon(P; x) = \inf_{\substack{x \in D \subset V \\ D \text{ an integral curve}}} \frac{P \cdot D}{\text{mult}_x D}.$$

The constants are independent of V , G , and P .

The upper estimate is the main geometric step. If a section can have large normalized order on a polarization of very small volume, its moving base locus produces smaller-dimensional covering members. Numerical subadjunction bounds their canonical degrees. Repeatedly passing to their Iitaka fibres will produce members of Kodaira dimension zero with very small section orders. If their chains fill V , the chain-order bound contradicts the initial large section order. Otherwise they give a proper general-type leaf covered by Kodaira-dimension-zero members, contradicting Lemma 6.4. Once the upper bound is proved, a thickening of a small-degree chain leaf gives the lower curve bound.

We first establish the local numerical input. For rational Cartier classes A, B , we write $A \preceq B$ when $B - A$ is rationally linearly equivalent to an effective rational divisor. A restriction of such a comparison will always be made to a member not contained in that effective divisor.

6.1 Numerical subadjunction and moving base components

Lemma 6.2 (Numerical subadjunction at a generic lc centre). *Let Z be a projective $\mathbb{Q}$ -factorial klt variety, let $\Theta \geq 0$ be a rational divisor, and let $S \subset Z$ be an integral subvariety of dimension $d > 0$. Suppose that (Z, Θ) is lc over a neighbourhood of the generic point of S , and that S is an lc centre there. For a nef rational Cartier divisor A on Z and a smooth projective resolution $S^* \rightarrow S$, one has*

$$K_{S^*} \cdot (A|_{S^*})^{d-1} \leq (K_Z + \Theta)|_S \cdot (A|_S)^{d-1}. \quad (6.2)$$

Here the right-hand side means the intersection of the restricted rational line bundle with the cycle S .

Proof. The arbitrary-boundary dlt blow-up theorem [15, Theorem 2.10] gives a projective birational morphism $p : Q \rightarrow Z$, with Q $\mathbb{Q}$ -factorial, such that

$$K_Q + B_Q + E = p^*(K_Z + \Theta), \quad (Q, B_Q) \text{ dlt}, \quad E \geq 0, \quad (6.3)$$

and $\text{Supp } E$ lies above the non-lc locus of (Z, Θ) . The theorem allows coefficients of Θ greater than one: B_Q is the truncated strict transform together with the reduced exceptional divisor, and the remaining crepant boundary is the effective excess E .

Choose a dlt stratum R minimal by inclusion among the strata mapping onto S . Such a stratum exists. Indeed, over the lc neighbourhood of the generic point of S , the morphism in (6.3) is crepant, and the centre on Q of an lc place with centre S is a dlt stratum mapping onto S . Dlt adjunction, in the form of the lc-centre and different theorem [27, 4.16], gives a normal R and an effective different. Adding the restriction of E gives an effective rational divisor Δ_R with

$$K_R + \Delta_R \sim_Q (K_Z + \Theta)|_R.$$

The restriction of E is defined and effective: E is rational Cartier and misses the generic fibre of $R \rightarrow S$. Moreover (R, Δ_R) is klt over the generic point of S . Any lc centre of the dlt different dominating S would give a smaller dlt stratum of Q dominating S , contrary to the choice of R ; the added excess is zero over that generic point.

Let S^ν be the normalization of S and factor the map through its normal Stein base:

$$R \xrightarrow{f} S' \xrightarrow{h} S^\nu \longrightarrow S, \quad f_*\mathcal{O}_R = \mathcal{O}_{S'}, \quad h \text{ finite.}$$

Write D for the pullback of $(K_Z + \Theta)|_S$ to S^ν . The morphism $f : (R, \Delta_R) \rightarrow S'$ satisfies the definition of a klt-trivial fibration in [14, Definition 3.1]. It is projective, generically subklt, and its adjoint is rationally linearly equivalent to f^*h^*D . The rank-one condition also holds: on a resolution

over the generic base point, the rounded discrepancy divisor is effective and exceptional over R . Allowing poles along such a divisor does not enlarge regular functions on the normal variety R , and connectedness of the fibres then gives rank one.

Ambro's canonical bundle formula, in the form recalled in [14, Theorem 3.5], therefore gives

$$h^*D \sim_{\mathbb{Q}} K_{S'} + B_{S'} + M_{S'}, \quad (6.4)$$

where $M_{S'}$ is the trace of b-nef rational data. The discriminant $B_{S'}$ is effective. To see this even where the total pair is not lc, let P be a prime of S' . Over its generic point, f^*P has a divisor of multiplicity at least one and Δ_R has nonnegative coefficient on that divisor. The corresponding threshold is therefore at most one. It can be negative, but its discriminant coefficient, one minus the threshold, remains nonnegative.

Let A' be the pullback to S' of $A|_{S^\vee}$. Intersect (6.4) with $(A')^{d-1}$. Both the effective discriminant and the b-nef moduli trace contribute nonnegatively. For the latter assertion, compute on a model where the moduli divisor is nef and apply the projection formula. If $e = \deg h$, codimension-one ramification for the finite map of normal varieties gives

$$K_{S'} \cdot (A')^{d-1} \geq e K_{S^\vee} \cdot (A|_{S^\vee})^{d-1}.$$

These are intersections of Weil divisors with Cartier powers, so the calculation can be made on the smooth codimension-one loci. Finally, the pushforward of K_{S^*} is $K_{S^\vee}$, and its exceptional divisors pair to zero with the pulled-back power of A . Dividing by e proves (6.2). $\square$

The moving-base-component strategy below adapts the jets construction in [34, Sections 7–8]. This method credit is distinct from the qualitative nonvanishing input in the introduction; the public jet-base-locus method and the present estimates are specified in the proof.

Lemma 6.3 (Tracking a moving base component). *Let Z be a projective $\mathbb{Q}$ -factorial klt variety of dimension $n \geq 2$, and let N be a big nef semiample rational Cartier divisor. Suppose there are $n + 1$ positive rational numbers, with consecutive spacing $\delta > 0$, all strictly between $(N^n)^{1/n}$ and $\gamma(N; Z)$. There is a covering family of integral subvarieties $S \subset Z$ of some dimension $0 < d < n$ such that, on a smooth projective resolution S^* of its geometric generic member,*

$$N^d \cdot S \leq (2/\delta)^{n-d} N^n, \quad (6.5)$$

$$K_{S^*} \cdot (N|_{S^*})^{d-1} \leq (K_Z + (2n/\delta)N)|_S \cdot (N|_S)^{d-1}. \quad (6.6)$$

Proof. The moving-parameter argument follows the jet-base-locus method of [12, Proposition 2.3]. We keep the subadjunction and degree estimates explicit for the rational polarizations used here. Choose a sufficiently divisible positive integer l , clearing the denominators of N and of the levels. It can be chosen arbitrarily large so that every jet space under consideration is nonzero: first choose a section witnessing an order above the highest level, and then take powers. At a variable very general smooth point x , write

$$V_\tau(x) = H^0(Z, \mathcal{O}_Z(lN) \otimes \mathfrak{m}_x^{l\tau}).$$

These spaces are the fibres of generic jet kernels, and thus have rationally varying bases after restricting the parameter space.

At each level the base locus through x is proper and has no isolated component at x . For the second assertion, if its local base ideal were primary to $\mathfrak{m}_x$, then n general members would

have local intersection multiplicity at least $(l\tau)^n$. Successive proper cuts, discarding other base components before each cut, would give

$$l^n N^n \geq (l\tau)^n,$$

contrary to $\tau > (N^n)^{1/n}$. This degree argument is also detailed below for the positive-dimensional component that we use.

The base loci are nested as the level increases. Choose successively an irreducible component through x , each contained in the next. Their dimensions are in $\{1, \dots, n-1\}$, so two consecutive components agree. Denote their common reduced component by S , and spread it over a parameter space containing the varying mark x . A finite extension of the parameter field may be used to name a geometric component. The resulting incidence dominates Z because it contains the graph of the mark.

Let τ and $\tau + \delta$ be the two levels. We claim that the base ideal of $V_{\tau+\delta}(x)$ has order at least $l\delta/2$ at the generic point of S . Differentiate a rationally varying basis of this jet space in parameter directions, keeping the spatial variable and a fixed basis of $H^0(Z, lN)$ constant. A derivative of length $r < l\delta$ has order at least $l(\tau + \delta) - r > l\tau$ at the varying mark, so it belongs to the lower jet space and vanishes along the incidence of S .

Here these derivative vanishings detect the order of the original section along the incidence. At its generic point, the ambient product is regular and both projections are dominant. Since the incidence dominates the spatial factor in characteristic zero, parameter derivations span the dual of its conormal space. The associated graded ring of the regular local ring is the symmetric algebra of that conormal space. A nonzero initial form of degree r is detected by some iterated parameter derivative of length r , by the characteristic-zero symmetric-power pairing. Thus all the vanishings just obtained force order at least $l\delta$, and in particular the asserted weaker bound. This is the same local Taylor argument used in Lemma 5.4. Passing to the geometric generic parameter fibre preserves it.

Put $a = n - d$. The local ring of Z at the generic point of S is regular of dimension a : a sweeping member meets the smooth open of Z . Since S is a component of the higher base locus, its base ideal there is primary to the maximal ideal $\mathfrak{m}$, and it is contained in $\mathfrak{m}^{[l\delta/2]}$. General choices of a members form a parameter ideal. Their intersection length, equal to its multiplicity in this regular local ring, is at least $(l\delta/2)^a$. Globally, take the cuts successively and discard components already contained in the base locus before the next cut. No discarded component contains the generic point of S , since S is itself an irreducible component of that base locus. The cuts are therefore proper at the generic point being tracked. Nefness of N makes all discarded degree contributions nonnegative, and gives

$$(l\delta/2)^a (N^d \cdot S) \leq l^a N^n.$$

This proves (6.5), as well as the isolated base-point calculation used above.

Let t be the local lc threshold of this base ideal over the generic point of S . A log resolution shows that t is positive and rational. Blowing up the smooth transverse centre, whose log discrepancy is a , gives

$$t \leq \frac{2a}{l\delta}.$$

For $a = 1$ the same inequality follows by testing S itself. Average sufficiently many general members of the higher jet space and multiply this average by t . On a log resolution of the ideal the free parts are general and their coefficients can be made arbitrarily small. This constructs an effective rational divisor

$$\Theta \sim_{\mathbb{Q}} bN, \quad b = lt \leq \frac{2a}{\delta},$$

such that (Z, Θ) is lc near the generic point of S and S is an lc centre there. Indeed a place computing the threshold has positive base-ideal order; the primary condition makes its centre equal to S after localization. Lemma 6.2 with $A = N$ now gives (6.6), since $a \leq n$. $\square$

We will also need the following elementary consequence of a covering family.

Lemma 6.4 (Canonical dimension of a covering member). *Let X be a projective variety birational to a smooth projective variety with a nonzero pluricanonical section. A smooth projective resolution of the geometric generic member of a covering family of integral subvarieties of X has nonnegative Kodaira dimension. If X is of general type, that resolution is of general type.*

Proof. Pass to a smooth projective model of X and to strict transforms of the moving members. Shrink the parameter space so that the incidence has fibres of pure constant dimension. If the incidence has dimension larger than $\dim X$, intersect the parameter space in a projective closure with general hyperplanes, in number equal to that excess. Over the generic point of X the incidence fibre embeds into the parameter space, so a general such section meets its top-dimensional part and makes the evaluation generically finite. A component of the cut incidence dominating X dominates a component of the parameter slice, by dimension.

Choose the cutting hyperplanes with independent generic coefficients over the original field of definition. A generic point of the selected parameter slice remains generic in the original parameter space over that field: every original proper closed subset has smaller dimension after the same general cuts. Thus the argument tests the original geometric generic member, rather than a special subfamily.

Resolve and compactify the cut incidence and its evaluation to X . The resulting generically finite morphism between smooth projective varieties satisfies

$$K_I = q^* K_X + R, \quad R \geq 0.$$

Restrict to a smooth geometric generic fibre of the parameter map. It is birational to the corresponding member, and adjunction identifies the restriction of K_I with its canonical divisor. A nonzero pluricanonical section of X pulls back to a section not identically zero on this generic fibre, by dominance. This proves nonnegative Kodaira dimension. If K_X is big, restrict a rational decomposition of a multiple into an ample class and an effective divisor. The ample pullback is big on the generic parameter fibre, since evaluation is generically finite on its image. Its canonical class is then big. $\square$

6.2 Small adjoint volumes on Iitaka fibres

The tracking lemma produces a moving member of dimension at most three whose polarization has small volume and whose canonical degree is bounded by that volume. The next lemma isolates the lower-dimensional argument: its Iitaka fibres have Kodaira dimension zero, and adding the restricted polarization to their canonical divisor still gives small volume.

Lemma 6.5. *Fix $1 \leq d \leq 3$ and a positive rational number C' . Let (W_i, L_i) be a sequence of smooth projective d -folds with $\kappa(W_i) \geq 0$ and big, nef, semiample rational Cartier divisors L_i . Suppose that*

$$L_i^d \longrightarrow 0, \quad K_{W_i} \cdot L_i^{d-1} \leq C' L_i^d. \quad (6.7)$$

After passing to a subsequence and resolving the Iitaka fibrations, their smooth geometric generic fibres U_i have a fixed positive dimension, satisfy $\kappa(U_i) = 0$, and obey

$$\text{vol}(K_{U_i} + L_i|_{U_i}) \longrightarrow 0.$$

Here L_i denotes its pullback to the resolved model; its restriction to U_i is big, nef, and semiample. When $\kappa(W_i) = 0$, the fibre U_i means W_i itself.

Proof. Suppress the sequence index. We first bound the volume of every fixed positive adjoint combination on W , then compare it with the volume on an Iitaka fibre. For every fixed positive rational number a , we claim that

$$\text{vol}(aK_W + L) \longrightarrow 0. \quad (6.8)$$

Choose a general divided member of a sufficiently divisible free multiple of L/a . It gives an effective klt boundary $\Gamma \sim_{\mathbb{Q}} L/a$. This boundary is big, and so is $K_W + \Gamma$, since $\kappa(W) \geq 0$. The klt log-general-type model and finite-generation theorem [2, Theorem 1.2] gives a $\mathbb{Q}$ -factorial klt log terminal model. Its nef big adjoint is semiample by the klt basepoint-free theorem [28, Theorem 3.3]: apply it to a Cartier multiple of the adjoint, whose sufficiently large multiple minus the adjoint is again nef and big.

On a common smooth resolution, let P_a be a times the pullback of this semiample adjoint. The negative-map comparison, using the negativity lemma [28, Lemma 3.39], gives

$$aK_W + L \sim_{\mathbb{Q}} P_a + F_a, \quad F_a \geq 0$$

after pullback, with F_a exceptional over the model. Sections in divisible degrees are unchanged by this exceptional part, so $P_a^d = \text{vol}(aK_W + L)$. All occurrences of L on the resolution mean its nef pullback. The Khovanskii–Teissier inequality gives

$$(aK_W + L) \cdot L^{d-1} \geq P_a \cdot L^{d-1} \geq (P_a^d)^{1/d} (L^d)^{(d-1)/d}.$$

For completeness, the nef mixed-intersection inequalities used here follow by ample perturbation and passage to the limit from the surface Hodge index theorem, applied on general complete-intersection surfaces. Combining with (6.7), we obtain the explicit bound

$$\text{vol}(aK_W + L) \leq (aC' + 1)^d L^d, \quad (6.9)$$

which proves the claim.

Pass to a subsequence on which $j = \kappa(W)$ is fixed. Since $d \leq 3$, effective Iitaka fibrations in the zero-boundary case give an integer $m_0 = m_0(d) > 0$ such that $|m_0 K_W|$ gives the Iitaka fibration; see [9, Corollary 1.5]. The case $j = d$ is impossible for large sequence index. Resolving that linear system would produce an integral free moving part with generically finite map and top self-intersection at least one. Thus $\text{vol}(K_W) \geq m_0^{-d}$, whereas $\text{vol}(K_W + L) \rightarrow 0$ by (6.8) and L has effective multiples.

If $0 < j < d$, resolve the Iitaka system, its Stein base, and the induced map to obtain a morphism of smooth projective varieties

$$f : W \longrightarrow T$$

with connected fibres. There is an integral big basepoint-free divisor H on T , pulled back from the hyperplane class of the system's image, such that

$$f^* H \preceq m_0 K_W, \quad H^j \geq 1. \quad (6.10)$$

We continue to write W for the resulting smooth model and L for its pullback. The previous intersection estimates persist: canonical divisors under further smooth resolutions acquire effective exceptional terms, and these pair to zero against L^{d-1} .

Let U be a smooth integral geometric generic fibre of f and put $g = d - j > 0$. Then $K_W|_U \sim K_U$, and $L|_U$ is big because these fibres move. We recall explicitly that $\kappa(U) = 0$. A nonzero canonical

multiple on W restricts nontrivially to the generic fibre, so $\kappa(U) \geq 0$. If a system $|kK_U|$ had positive-dimensional image, its sections would extend over the generic base after twisting kK_W by $v f^* H$ for a sufficiently large v . Indeed the coherent direct image of kK_W has the required generic fibre of sections, and a sufficiently large multiple of the big divisor H makes it generically generated, using a decomposition into ample and effective parts. By (6.10), the twist is absorbed into a further canonical multiple. The resulting section ratio is nonconstant on the geometric generic fibre and hence transcendental over $\mathbb{C}(T)$. Products with sections of $|m_0 K_W|$ also preserve the j independent base ratios. This would force $\kappa(W) > j$, a contradiction. When $j = 0$, simply put $U = W$ and $g = d$.

We next show that

$$\text{vol}(K_U + L|_U) \longrightarrow 0. \quad (6.11)$$

For $j = 0$ this is (6.8). If $j > 0$ and $g = 1$, then $\kappa(U) = 0$ implies $\deg K_U = 0$, and the volume in question equals $\deg(L|_U)$. Set $A = f^* H$. Nef mixed-intersection log-concavity and (6.10) give

$$A^j L^{d-j} \leq \left(\frac{A L^{d-1}}{L^d} \right)^j L^d \leq (m_0 C')^j L^d \longrightarrow 0.$$

One may add a positive multiple of L to A and pass to the limit if a mixed intersection vanishes. The projection formula and $H^j \geq 1$ show that this bounds $\deg(L|_U)$ from above.

The only remaining case is $d = 3$, $j = 1$, and $g = 2$. Now T is a smooth curve and H is ample. A basepoint-free pencil in $|H|$ and its ramification divisor give $K_T + 2H \succeq 0$. Consequently

$$K_{W/T} + L + 2f^* H \preceq (1 + 4m_0)K_W + L. \quad (6.12)$$

Choose an effective klt rational boundary linearly equivalent to L from its semiample system. The twisted weak positivity theorem [16, Theorem 1.1] applies to f with this boundary. Thus for sufficiently divisible l the locally free sheaf

$$E_l = f_* \mathcal{O}_W(l(K_{W/T} + L))$$

is weakly positive. Here smoothness, projectivity, connected fibres, and effectivity of the klt boundary are all satisfied; on the smooth curve a torsion-free direct image is locally free.

Choose $l \geq 1$, separately for each member of the sequence, so that the l -th Veronese of the section algebra of $K_U + L|_U$ is generated in degree one. Such an l exists by the same big klt adjoint model and finite-generation theorem used above. Enlarge it to clear the chosen rational linear equivalence. Weak positivity gives $b > 0$ for which

$$\text{Sym}^b E_l \otimes \mathcal{O}_T(bH)$$

is generated by global sections at the generic point. For every positive multiple q of b , symmetric multiplication and the degree-one generation on the generic fibre therefore give global sections of

$$ql(K_{W/T} + L) + qf^* H$$

spanning the degree- ql fibre sections. Select a basis over $\mathbb{C}(T)$ among their restrictions, of cardinality $h^0(U, ql(K_U + L|_U))$. Multiplying by a basis of $H^0(T, qlH)$ gives linearly independent products over $\mathbb{C}$: any relation can first be tested using independence over $\mathbb{C}(T)$ and then independence of the base sections. Multiplication by a nonzero section of $(ql - q)H$ places these products in degree ql of $K_{W/T} + L + 2f^* H$.

Curve Riemann–Roch and the asymptotic fibre section formula, with l fixed and $q \rightarrow \infty$ through these multiples, yield

$$\begin{aligned} \operatorname{vol}((1 + 4m_0)K_W + L) &\geq \operatorname{vol}(K_{W/T} + L + 2f^*H) \\ &\geq \frac{d!}{g!} \deg(H) \operatorname{vol}(K_U + L|_U). \end{aligned} \quad (6.13)$$

The first expression tends to zero by (6.8), and $\deg H \geq 1$. This proves (6.11) in every case. $\square$

6.3 The upper order estimate

Proof of the upper bound in Theorem 6.1. If no uniform C exists, rational rescaling gives a sequence of the data in the theorem, with polarizations denoted M , such that

$$M^4 \longrightarrow 0, \quad \gamma(M; V) > 1. \quad (6.14)$$

The proof decreases the dimension while retaining small volume and an effective canonical upper bound. At each stage we first extract a moving base component, then pass to a Kodaira-dimension-zero fibre of its Iitaka fibration, and finally replace a big adjoint by its semiample model. The process stops when its section-order invariant tends to zero. Tracing the effective divisor comparisons back will construct covering families of positive-dimensional proper subvarieties $A \subset V$ whose geometric generic members satisfy

$$\kappa(A^*) = 0, \quad \gamma(M; A) \longrightarrow 0. \quad (6.15)$$

All constructions can be made successively on geometric generic members. To be precise, choose countable fields of definition for the sequence. A geometric generic datum over such a field can be realized over $\mathbb{C}$ to apply the model and positivity theorems used here. The resulting finite collection of varieties, morphisms, and divisors descends again to a field finitely generated over the preceding generic parameter field. Spreading it out therefore gives a dominant extension of the preceding parameter space: the old generic parameter is never specialized. Formation of section spaces and of their jet kernels commutes with field extension, so the generic all-degree order comparisons are preserved. There are only finitely many stages for each variety, and countably many varieties in the sequence.

Begin with a small projective $\mathbb{Q}$ -factorialization $Z \rightarrow V$, which exists for the klt variety V by [2, Corollary 1.4.3], and let N be the pullback of M to Z . We describe a stage of the construction. After passing to a subsequence, its dimension n is fixed and the following properties hold:

- (i) Z is projective $\mathbb{Q}$ -factorial klt and N is big, nef, and semiample, with $N^n \rightarrow 0$;
- (ii) a smooth projective model of Z has nonnegative Kodaira dimension;
- (iii) $K_Z \preceq a_0 N$ for a fixed positive rational number a_0 ;
- (iv) $\gamma(N; Z)$ is bounded below by a positive constant.

Initially these assertions follow from (6.14), canonicity, and crepantness of the small morphism. If $n = 1$, they cannot all hold, since the order of a section on a smooth curve is bounded by its degree. We may therefore assume $n \geq 2$.

Choose $n + 1$ fixed rational jet levels, equally spaced by $\delta > 0$, below the lower bound in (iv). For all sufficiently large sequence indices they are above $(N^n)^{1/n}$. Apply Lemma 6.3. Write $W = S^*$

for a resolution of its geometric generic moving member and $L = N|_W$, and pass to a subsequence with fixed $d = \dim W < n$. The divisor L is nef and semiample, and is big because a sweeping member meets the open set on which the semiample map of N is generically finite. Lemma 6.4 gives $\kappa(W) \geq 0$. Restricting the effective comparison in (iii) to the moving member and using the tracking bounds yields

$$L^d \longrightarrow 0, \quad K_W \cdot L^{d-1} \leq C' L^d \quad (6.16)$$

with the fixed constant $C' = a_0 + 2n/\delta$ at this stage.

Apply Lemma 6.5 to these moving members. After passing to a subsequence and resolving their Iitaka systems, let U be the smooth geometric generic Iitaka fibre, with $U = W$ when $\kappa(W) = 0$, and put $g = \dim U$. Then

$$0 < g \leq d < n, \quad \kappa(U) = 0, \quad \text{vol}(K_U + L|_U) \longrightarrow 0.$$

The restricted divisor $L|_U$ is big, nef, and semiample. The generic field and spreading convention above keeps these fibres in covering families over the preceding parameter space.

To continue the dimension descent, we need a semiample polarization that still bounds the original restricted sections from above. Take an effective klt boundary $\Gamma \sim_{\mathbb{Q}} L|_U$ from the free system, and take a big klt good model of $K_U + \Gamma$, as in the proof of Lemma 6.5. Write Z' for its $\mathbb{Q}$ -factorial klt underlying variety and N' for its big semiample adjoint. Then

$$(N')^g = \text{vol}(K_U + L|_U) \longrightarrow 0.$$

On a smooth common resolution with maps a to U and b to Z' , the negative-map comparison reads

$$a^*(K_U + \Gamma) = b^*N' + F, \quad F \geq 0 \text{ exceptional over } Z'. \quad (6.17)$$

For any effective $\Delta_U \sim_{\mathbb{Q}} K_U$, we have

$$F \leq a^*\Delta_U. \quad (6.18)$$

Indeed, for sufficiently divisible m , every effective divisor in $|mb^*N' + mF|$ contains mF . Its pushforward is an effective divisor in $|mN'|$ on the normal variety Z' , and pulling that Cartier divisor back leaves exactly the exceptional summand mF . But $ma^*\Delta_U + Y$ belongs to this system when Y is a general member of the free system $|ma^*(L|_U)|$. Choose Y avoiding every component of F . Comparing coefficients proves (6.18). In particular,

$$a^*(L|_U) \preceq b^*N'. \quad (6.19)$$

If $\gamma(N'; Z') \rightarrow 0$ on a subsequence, we stop. Otherwise, on a subsequence it is bounded below by a positive constant, and we repeat with (Z', N') . The four stage conditions hold: $K_{Z'} \preceq N'$ because the pushed-forward boundary is effective, and a smooth model of Z' is birational to U , which has Kodaira dimension zero. The dimension has decreased strictly, from n to $g \leq d < n$. The construction must stop, since on a curve maximum normalized section order is at most the degree.

We verify that the stopping condition proves (6.15) for the original polarization. Every moving member and fibre used above spreads to a covering family over the preceding generic parameter space. Its map to its image in that space, and ultimately in V , is birational: a moving member meets the open set of isomorphism of each relevant birational model map. For a subsequent member in Z' , take its strict transform on the common resolution, so it has an actual morphism back to U . The effective comparisons (6.19) restrict to these members, since they are not contained in the extra divisors. Their successive restrictions compare the pullback of the original M with the final N' on a common resolution. Multiplication by the effective differences injects section spaces in divisible

degrees, and at a generic marked point those differences do not vanish. Birational pullback between normal projective models preserves the section spaces. Thus the final image $A \subset V$ satisfies

$$\gamma(M; A) \leq \gamma(N'; Z') \longrightarrow 0, \quad \kappa(A^*) = \kappa(U) = 0.$$

Composing the dominant incidence evaluations proves that these images form a covering family. They have positive dimension and are proper because the first tracking step already decreased dimension.

Mark a generic smooth point on these members and add every G -translate of the marked family. All data can be defined over one countable field; invariance of M preserves the order bound. Lemma 5.1 gives a G -equivariant rational quotient, and Lemma 5.4 bounds the section order on its geometric generic leaf by its dimension times B , where $B \rightarrow 0$. If the leaf has dimension four, this gives $\gamma(M; V) \leq 4B$, contradicting (6.14). Otherwise it gives a G -equivariant rational fibration with positive-dimensional base and fibre. The geometric generic leaf is of general type by hypothesis.

That leaf is nevertheless covered by members of Kodaira dimension zero. To see that the assertion is about generic members, restrict the marked incidence to its generic leaf name. A component still dominates the leaf by marked-point evaluation. Its generic parameter is generic in the original parameter space over the original field, because it is obtained by geometric extension over the generic name. The quotient is generically constant on each such member, and its strict transforms give a covering family on a resolution of the leaf. This contradicts Lemma 6.4. The upper bound follows. $\square$

6.4 The lower curve estimate

Proof of the lower bound in Theorem 6.1. Rescale rationally so that $1 \leq P^4 \leq 2$. If no uniform positive lower bound exists, there is a sequence of marked covering families of integral curves with

$$\frac{P \cdot D}{\text{mult}_x D} \leq B, \quad B \longrightarrow 0.$$

Here the marks can be chosen generic over countable fields of definition. Indeed, failure of a very general bound still gives a bad mark after excluding the countably many proper closed subsets over such a field. Choose a curve witnessing a strict upper bound there and spread the curve and its mark over their field of definition. The mark evaluation is dominant, and degree and multiplicity agree with those of the geometric generic data. Include all G -translates if necessary; they obey the same ratio bound.

Take the rational quotient from Lemma 5.1. After passing to a subsequence with fixed leaf dimension $h > 0$, Lemma 5.4 gives, on its geometric generic leaf H ,

$$P^h \cdot H \leq (hB)^h. \tag{6.20}$$

For large sequence index it cannot have dimension four, since $P^4 \geq 1$. Put $s = 4 - h > 0$. Resolve the leaf fibration to a morphism $f : X \rightarrow T$ between smooth projective varieties with geometrically integral generic fibre, and let $\pi : X \rightarrow V$ be the resulting birational morphism. Continue to write P for π^*P .

Choose a smooth integral very general fibre F of f . It can be chosen not contained in any of the countably many proper closed sets excluded by the all-degree upper order bound, nor in the exceptional locus of π . For clarity, over a smooth open of the base, each proper closed set of X contains the whole fibre only over a proper closed subset of T , by properness and upper

semicontinuity of fibre dimension. Exclude these countably many subsets first, then choose a point $x \in F$ outside their intersections with F . Thus π is an isomorphism near x and

$$\gamma(P; V) \leq C2^{1/4} \quad (6.21)$$

holds at $\pi(x)$ in every divisible degree. The degree $P^h \cdot F$ equals the geometric generic degree in (6.20).

Let $I = I_F \subset \mathcal{O}_X$. Since F is a smooth fibre over a smooth point of the s -dimensional base, it is regularly embedded and has trivial conormal bundle of rank s . Consequently

$$I^k / I^{k+1} \simeq \text{Sym}^k(\mathcal{O}_F^{\oplus s}) \simeq \mathcal{O}_F^{\oplus \binom{k+s-1}{s-1}}.$$

The filtration by powers of I therefore bounds the rank of restriction to the u -th thickening, for every $u \geq 1$ and every divisible l , by

$$h^0(X, \mathcal{O}_X(lP) \otimes (\mathcal{O}_X / I^u)) \leq \binom{u+s-1}{s} h^0(F, lP|_F). \quad (6.22)$$

No vanishing theorem uniform in u is required for this inequality.

Fix $D > C2^{1/4}$ and set $u = \lceil Dl \rceil$. The pullback of P is big and semiample on X , and its restriction to F is big and semiample because F meets the birational isomorphism locus. The semiample Hilbert-polynomial asymptotics show that the leading coefficient, after division by l^4 , on the right-hand side of (6.22) is

$$\frac{D^s(P^h \cdot F)}{s! h!} \leq \frac{D^s(hB)^h}{s! h!}.$$

Choose the sequence index first so that this is strictly less than $1/4!$, hence less than $P^4/4!$. Then choose a sufficiently large divisible l for this fixed example. Since

$$h^0(X, lP) = \frac{P^4}{4!} l^4 + O(l^3),$$

restriction to the thickening has a nonzero kernel. A kernel section lies in I^u , and $I_x^u \subset \mathfrak{m}_x^u$. It therefore has order at least $u \geq Dl$ at x . Normality gives $\pi_* \mathcal{O}_X = \mathcal{O}_V$, so it is a section downstairs as well, with the same order at the point where π is an isomorphism. This contradicts (6.21). Rescaling back proves the lower bound, and completes the theorem. $\square$

7 Quadratic jet cost for diagonal cyclic products

We prove a bound for diagonal products of the canonical cyclic cover. The constant in the bound depends only on an upper bound for normalized section orders on the original fourfold. In particular, it is independent of the order of the cover.

The diagonal rank-deficit and determinant-order strategy adapts the Frobenius construction of [34, Section 9]. The cyclic many-factor weights, estimates, and constants needed here are supplied below; the abundance conclusion of that reference is not an input to this argument.

Proposition 7.1. *Let X be a normal projective canonical fourfold over $\mathbb{C}$ with $K_X \sim_{\mathbb{Q}} 0$, and let r be the least positive integer such that $rK_X \sim 0$. Let $\pi : Y \rightarrow X$ be its canonical cyclic cover, with group μ_r . Fix $C_1 \in \mathbb{Q}_{>0}$, and choose $b_0 \in \mathbb{Q}_{>0}$ satisfying*

$$3(2b_0)^4 < \frac{1}{4}, \quad b_0(C_1 + 1) < \frac{3}{4}. \quad (7.1)$$

Suppose that L is an ample rational Cartier divisor on X such that

$$1 \leq L^4 \leq 2, \quad \gamma(L; X) \leq C_1,$$

where γ is as in Definition 5.3. For each integer $t \geq 2$, set

$$Z_t = Y^t / \mu_{r, \text{diag}}, \quad \theta_t : Z_t \longrightarrow X^t, \quad P_t = \theta_t^* \left(\sum_{i=1}^t \text{pr}_i^* L \right).$$

Then P_t is ample and, at a very general point z of Z_t ,

$$\varepsilon(P_t; z) \leq \frac{5t^2}{b_0}. \quad (7.2)$$

Suppose the asserted curve bound fails. The resulting surplus of ordinary jets will produce, after reduction to characteristic p , a map of vector bundles that has full rank at a general product point. On the diagonal its rank is at most one quarter of the target rank. A nonzero maximal minor must consequently vanish to high order there. The section-order bound downstairs will give a strictly smaller upper bound for that same minor at one diagonal point.

The next three lemmas supply the estimates in this comparison. The truncated-power slope bound controls the graded pieces on the Frobenius diagonal. The fixed flag converts those slopes into a bound for their spaces of sections, hence for the diagonal rank. The external-product order lemma bounds the minor by the sum of its slot orders and rules out a larger order caused by cancellation. We first establish these estimates on vector bundles. For a vector bundle E on a smooth projective curve, write $\mu_{\min}(E)$ and $\mu_{\max}(E)$ for the smallest and largest slopes in its Harder–Narasimhan filtration.

Lemma 7.2. *Let C be a smooth projective integral curve of genus g over an algebraically closed field k of characteristic $p > 0$. Let V be a vector bundle of rank n with $\mu_{\min}(V) \geq 0$, and put*

$$G = \max\{0, 2g - 2\}, \quad u = n(p - 1), \quad c_2 = n(n - 1)G.$$

Let $A_\bullet(V)$ denote the symmetric algebra of V modulo the ideal locally generated by the p -th powers of its linear generators. Thus, in a local frame, $A_\bullet(V)$ is the graded algebra $\mathcal{O}_C[v_1, \dots, v_n] / (v_1^p, \dots, v_n^p)$. Its graded pieces are locally free, it is zero in degrees above u , and

$$\mu_{\max}(A_j(V)) \leq (p - 1) \deg(V) + c_2 \quad (0 \leq j \leq u). \quad (7.3)$$

Proof. We use the canonical-connection approach to Frobenius instability; see Langer [29, §2, Corollaries 2.4 and 6.2]. We give the estimate and tensor argument explicitly to retain the constants needed for the many-factor diagonal. Let F_C denote absolute Frobenius. For every vector bundle U of rank n , we first claim that

$$\mu_{\min}(F_C^* U) \geq p\mu_{\min}(U) - (n - 1)G. \quad (7.4)$$

The pullback $F_C^* U$ carries the connection given by componentwise differentiation in pulled-back frames: the transition matrices have p -th-power entries. Write the strictly decreasing slopes of its Harder–Narasimhan factors as $\nu_1 > \dots > \nu_a$. If some successive gap is greater than G , take the last such gap, and let S be the saturated filtration step immediately before that gap. Then

$$\mu_{\min}(S) > \mu_{\max}((F_C^* U)/S) + \deg(\Omega_C^1).$$

Consequently the second fundamental map

$$S \longrightarrow ((F_C^*U)/S) \otimes \Omega_C^1$$

vanishes, so S is preserved by the connection.

Here this invariance gives Frobenius descent directly. At the function field $K = k(C)$, choose a matrix whose columns form a basis of the subspace S_K and whose rows in a suitable index set form the identity matrix. Differentiating its columns gives vectors in $S_K \otimes \Omega_{K/k}^1$. Their identity rows are zero, so these differentiated columns are all zero. Every matrix entry therefore lies in K^p . Indeed $[K : K^p] = p$ for a one-variable function field over a perfect field, and the nonzero derivation $d : K \rightarrow \Omega_{K/k}^1$ has kernel K^p . Extracting roots of the matrix entries produces a subspace of U_K . Let $S_0 \subset U$ be its saturation. Since C is smooth, U/S_0 is locally free. Frobenius is flat, so $F_C^*S_0$ is saturated in F_C^*U . It has the same generic fibre as S , hence equals S .

Put $Q = (F_C^*U)/S$. By descent it is the Frobenius pullback of a locally free quotient of U , and therefore $\mu(Q) \geq p\mu_{\min}(U)$. By the choice of the last large gap, all successive Harder–Narasimhan slope gaps of Q are at most G . Its average slope exceeds its smallest slope by at most $(n-1)G$. Moreover $\mu_{\min}(Q) = \mu_{\min}(F_C^*U)$. These observations prove (7.4). If there is no gap greater than G , the same argument applies with $S = 0$ and $Q = F_C^*U$.

Iteration, using $\mu_{\min}(V) \geq 0$, gives for every $e \geq 1$

$$\mu_{\min}((F_C^e)^*V) \geq -(n-1)G \frac{p^e - 1}{p - 1}. \quad (7.5)$$

We next deduce a tensor estimate without assuming tensor semistability in characteristic p . For each e , choose a line bundle T_e of integral degree d_e such that

$$2g - \mu_{\min}((F_C^e)^*V) \leq d_e < 2g + 1 - \mu_{\min}((F_C^e)^*V).$$

The bundle $V_e = (F_C^e)^*V \otimes T_e$ is globally generated. In fact, after subtracting any point its minimum slope is at least $2g - 1$, so Serre duality and the slope criterion for a nonzero homomorphism give $H^1(C, V_e(-x)) = 0$; evaluation at x is therefore surjective.

Let Q_i be any positive-rank torsion-free quotient of $V^{\otimes i}$. Its pullback, twisted by $T_e^{\otimes i}$, is a quotient of $V_e^{\otimes i}$, hence is globally generated and has nonnegative degree. Thus

$$p^e \mu(Q_i) + i d_e \geq 0.$$

Divide by p^e and let e tend to infinity. The upper bound for d_e and (7.5) imply

$$\mu(Q_i) \geq -\frac{i(n-1)G}{p-1}. \quad (7.6)$$

Since $A_i(V)$ is a quotient of $V^{\otimes i}$, the same lower bound holds for its minimum slope. For $0 \leq i \leq u$, it is at least $-c_2$.

Multiplication in the truncated algebra is a perfect pairing

$$A_j(V) \otimes A_{u-j}(V) \longrightarrow A_u(V).$$

In a frame, each monomial pairs with its unique complementary exponent monomial, with coefficient one. The top line has transition character $(\det V)^{p-1}$: this follows for diagonal changes of frame and for elementary changes of frame, which generate the general linear group. It follows that

$$A_j(V)^* \otimes (\det V)^{p-1} \simeq A_{u-j}(V).$$

Its minimum slope is at least $-c_2$. Slope duality now gives (7.3). $\square$

Lemma 7.3. *Let W be a smooth projective fourfold over an algebraically closed field, and let H be an ample rational Cartier divisor. Fix a smooth complete intersection flag, cut successively by divisors from $|l_1 H|$, $|l_2 H|$, and $|l_3 H|$, ending in a smooth integral curve C ; here each $l_i H$ is an integral ample divisor. Write $d_i = l_i H \cdot C$. If E is a vector bundle of rank e and $\mu_{\max}(E|_C) \leq M$ for a real number $M \geq 0$, then*

$$h^0(W, E) \leq e(M+1) \prod_{i=1}^3 \left(1 + \frac{M}{d_i}\right). \quad (7.7)$$

Proof. Successive applications of the divisor exact sequences along the flag bound $h^0(W, E)$ by

$$\sum_{k_1, k_2, k_3 \geq 0} h^0 \left(C, E|_C \otimes \mathcal{O}_C \left(- \sum_{i=1}^3 k_i l_i H \right) \right). \quad (7.8)$$

To justify the iterations, at each flag stage and with the preceding twists fixed, sufficiently negative ample twists have no sections. A vector bundle on that stage embeds into a finite direct sum of copies of a sufficiently positive ample line bundle, by dualizing a global generation map for its dual. This gives the required vanishing and allows each iteration of the exact sequence to terminate. Enlarging the resulting finite sums to (7.8) can only increase the bound.

A summand vanishes when $\sum_i k_i d_i > M$, because its maximum slope is then negative. Each summand is at most $e(M+1)$: evaluation at $\lfloor M \rfloor + 1$ distinct points is injective, since subtracting these points makes the maximum slope negative even before the additional nonpositive twists. There are at most $\prod_i (1 + M/d_i)$ possible triples with $\sum_i k_i d_i \leq M$. This proves (7.7). $\square$

Lemma 7.4. *For $1 \leq i \leq t$, let V_i be an integral projective variety over an algebraically closed field, let $x_i \in V_i$ be a smooth point, and let M_i be a line bundle with $H^0(V_i, M_i) \neq 0$. Put*

$$a_i = \max_{0 \neq s \in H^0(V_i, M_i)} \text{ord}_{x_i}(s).$$

Every nonzero section s of the external product $\otimes_i \text{pr}_i^ M_i$ satisfies*

$$\text{ord}_{(x_1, \dots, x_t)}(s) \leq \sum_{i=1}^t a_i.$$

Proof. Each space of sections is finite dimensional, and its order filtration is separated; the numbers a_i are therefore finite. Choose a basis adapted to this filtration in each slot. The initial forms of basis elements having a fixed order are linearly independent in that degree of the associated graded local ring. By the Künneth formula, every section of the external product is a linear combination of the tensor products of these bases. In the least total degree occurring in a nonzero combination, group its initial forms by their slot multidegrees. Different multidegrees cannot cancel. Within each multidegree the products are independent, since they are tensor products of independent initial forms in disjoint sets of smooth parameters. The section's order is consequently the least total degree of one of its nonzero basis coefficients, and this is at most $\sum_i a_i$. $\square$

Proof of Proposition 7.1. The map θ_t is finite, so P_t is ample. Fix X, L, r , and t ; all auxiliary data below may depend on these choices. Set

$$b = \frac{b_0}{t}, \quad q = 2tr. \quad (7.9)$$

Suppose, for a contradiction, that $\varepsilon(P_t; z) > 5t/b$ at a smooth very general point z , whose image in every X -slot lies in X_{reg} .

Jets and the canonical torsor. Choose a rational number c with

$$5t < c < b\varepsilon(P_t; z).$$

On the blow-up $\sigma : \tilde{Z}_t \rightarrow Z_t$ at z , with exceptional divisor E_z , the rational divisor $\sigma^*(bP_t) - cE_z$ is ample. To see this, choose a larger coefficient below $b\varepsilon(P_t; z)$ for which the corresponding divisor is nef, and interpolate it with the ample divisor obtained at a sufficiently small positive coefficient. Serre vanishing, applied to sufficiently large divisible multiples of $\sigma^*(bP_t) - cE_z$, makes the restriction map to the corresponding thick exceptional divisor surjective. Pullback identifies sections of k_0bP_t with sections of its pullback, by normality of Z_t . At the smooth point z , ordinary jets of order strictly less than k_0c inject into the sections on k_0cE_z . It follows, after choosing k_0 large and divisible, that

$$\begin{aligned} Q = k_0bL \text{ is integral Cartier,} \\ H^0(Z_t, k_0bP_t) \longrightarrow \mathcal{O}_{Z_t, z} / \mathfrak{m}_z^{5tk_0+1} \text{ is surjective,} \end{aligned} \tag{7.10}$$

where a local frame is understood in the displayed evaluation map.

Fix a section η trivializing rK_X . Over X_{reg} , the cover $Y \rightarrow X$ is the torsor of frames τ of the canonical line with $\tau^r = \eta$. Over tuples in this open locus the diagonal action is free, and $Y^t \rightarrow Z_t$ is étale. Choose a lift $\tilde{z}$ of z . Pulling back (7.10) preserves jet surjectivity at $\tilde{z}$. The pulled-back sections are diagonal invariants. By the Künneth formula and decomposition into characters, they are linear combinations of pure tensors whose slot factors have the form

$$\begin{aligned} \tau^{-m_i} s_i, \quad s_i \in H^0(X_{\text{reg}}, \mathcal{O}(Q + m_i K_X)), \\ 0 \leq m_i < r, \quad \sum_i m_i \equiv 0 \pmod{r}. \end{aligned} \tag{7.11}$$

Indeed multiplication of an eigenvector by the corresponding power of the tautological frame makes it invariant and hence descending to the indicated line bundle.

Take a smooth projective resolution $u : W \rightarrow X$, isomorphic over X_{reg} , and write

$$K_W = u^*K_X + A, \quad A \geq 0 \text{ exceptional.}$$

We use the same letters L, Q for their pullbacks to W . Every s_i in (7.11) extends to a section of $Q + m_i K_W$. In fact, its effective zero divisor extends by closure on the normal variety X ; this divisor is rational Cartier, since its class is $Q + m_i K_X$. Pulling it back and adding $m_i A$ gives the effective divisor of the corresponding rational section on W . Similarly η extends to

$$\eta_W \in H^0(W, rK_W), \quad \text{Div}(\eta_W) = rA.$$

Let $U = u^{-1}(X_{\text{reg}})$. On the torsor of roots of η_W over U , a fixed finite list of the tensors (7.11), now using these sections on W , spans the jets in (7.10) at the chosen lift.

A polarization, a flag, and a movable test. Fix an ample divisor H_{amp} on W . Since L is nef and $K_W L^3 = 0$, a sufficiently small positive rational δ makes $H = L + \delta H_{\text{amp}}$ ample and ensures

$$H^4 \leq 3, \quad LH^3 \leq H^4, \quad (q + t - 1)K_W H^3 \leq b_0 H^4. \tag{7.12}$$

The equality $K_W L^3 = 0$ follows from $K_X \sim_{\mathbb{Q}} 0$ and the exceptionality of A . The canonical class of W is pseudoeffective. By [7, Theorem 2.1], Ω_W^1 is generically semipositive for this polarization. Apply [31, Theorem 6.1] to its finitely many Harder–Narasimhan factors. We can choose a smooth

complete intersection flag, cut successively by sufficiently positive integral multiples $l_i H$, $1 \leq i \leq 3$, ending in a smooth integral curve C such that

$$\mu_{\min}(\Omega_W^1|_C) \geq 0. \quad (7.13)$$

The general curve avoids the codimension-two loci where the factors or their filtration quotients fail to be locally free. Their restricted slopes are the original slopes multiplied by the same positive factor $l_1 l_2 l_3$, so the restriction theorem gives (7.13).

Choose also a very general point $x \in U$ at which the all-degree bound defining $\gamma(L; X) \leq C_1$ holds. Let $\pi_x : \widehat{W} \rightarrow W$ be its blow-up, with exceptional divisor J , and put

$$D^+ = bL + (q+1)K_W, \quad d_0 = b(C_1 + 1). \quad (7.14)$$

The divisor D^+ is big. In sufficiently divisible degrees its section spaces agree with those of bL , after multiplication by a fixed section whose zero divisor is exceptional over X . This is because $rK_X \sim 0$, A is effective and exceptional, and $u_* \mathcal{O}_W(E) = \mathcal{O}_X$ for any effective integral exceptional divisor E on this resolution. At x these fixed sections have order zero. Thus every nonzero section of a sufficiently divisible multiple mD^+ has order at most mbC_1 at x .

It follows that $\pi_x^* D^+ - d_0 J$ is not pseudoeffective. Otherwise, interpolating it with the big divisor $\pi_x^* D^+$ would make $\pi_x^* D^+ - dJ$ big for a rational d with $bC_1 < d < d_0$. An effective sufficiently divisible multiple would give a section of a multiple of D^+ with an excluded order at x . The projective algebraic duality theorem of [6, Theorem 2.2, arXiv version] therefore supplies a strongly movable class β on $\widehat{W}$ such that

$$D^+ \cdot \beta < d_0 J \cdot \beta, \quad J \cdot \beta > 0. \quad (7.15)$$

Here and subsequently divisors on W are pulled back when paired with β . More precisely, we may take $\beta = v_*(A_1 A_2 A_3)$ for a smooth projective birational model $v : T \rightarrow \widehat{W}$ and very ample divisors A_i on T . The dual cone is the closure of the cone generated by such classes; strict negativity on that cone yields negativity on one generator. Also $D^+ \cdot \beta \geq 0$, since L is nef and K_W is pseudoeffective. The second inequality in (7.15) follows from the first and $d_0 > 0$. In particular,

$$L \cdot \beta \geq 0, \quad K_W \cdot \beta \geq 0. \quad (7.16)$$

Reduction of the fixed data. Spread the preceding finite algebraic data over an integral finitely generated $\mathbb{Z}$ -subalgebra of $\mathbb{C}$. This includes the smooth varieties, the flag, the points and the blow-up, the morphism v and its very ample divisors, suitable integral multiples of the rational line bundles, the sections used in (7.11), the section η_W , and the torsor and its marked lift over U . After shrinking the base, the relevant fibres of W, C, T , and $\widehat{W}$ are smooth, projective, and geometrically integral; the testing morphism remains dominant; and all required ampleness and intersection data persist. The fixed finite jet evaluation remains surjective: it is a map of finite locally free modules obtained by restriction to a fixed-order infinitesimal neighbourhood of a section of a smooth morphism, and surjectivity is open. Invert r and the finitely many denominators used in the data.

We can also retain (7.13) on every geometric fibre of this smaller base. To see this without imposing any Frobenius semistability condition, choose a fixed relative ample twist that makes the family $\Omega_W^1|_C$ relatively globally generated. For every positive-rank quotient, its degree is then bounded below uniformly, in terms of its rank and this twist. Hence there are only finitely many possible negative degrees and ranks, and therefore finitely many Hilbert polynomials, for quotients of negative degree. The corresponding relative Quot schemes are projective. None has a point on

the geometric generic fibre: such a point would persist after extension to the original complex field and contradict (7.13). This also excludes quotients with torsion and negative degree, since their torsion-free quotients have still smaller degree. Remove the images of these finitely many Quot schemes. The remaining geometric fibres have $\mu_{\min}(\Omega_W^1|_C) \geq 0$.

There are geometric fibres in arbitrarily large characteristics p : the base is nonempty, of finite type, and dominant over $\text{Spec } \mathbb{Z}$. We take such fibres over algebraic closures of residue fields and keep the preceding notation. The genus of C and all the intersection numbers used above are constant. The class β continues to pair nonnegatively with every effective divisor: pull the divisor back under the dominant morphism v and intersect with $A_1 A_2 A_3$. This argument applies also after a base-field twist. Thus only the fixed test, and not pseudoeffective-cone duality, is being specialized.

Full rank away from the diagonal. In one of these reductions, write $F : W \rightarrow W'$ for relative Frobenius, where a prime denotes base-field twist. It is finite flat of degree p^4 . Put

$$\begin{aligned} N &= \lfloor p/k_0 \rfloor, & B &= NQ, & \mathcal{E} &= F_* \mathcal{O}_W(B), \\ s &= p^4 = \text{rk}(\mathcal{E}), & R &= s^t. \end{aligned} \tag{7.17}$$

Products of N of the fixed jet tensors span jets through degree $5tk_0N$ at the chosen lift. Indeed every monomial of degree at most $5tk_0N$ is a product of N monomials of degree at most $5tk_0$. Choose representatives with these leading terms from the original jet-spanning space. Their products give a triangular spanning set for the jet filtration, in every characteristic. For all sufficiently large p ,

$$5tk_0N \geq 4t(p-1).$$

The local monomial inclusion used here is the one in [32, proof of Proposition 2.12, equation (2.8)], with smooth dimension $4t$. The products therefore span on the thick point defined by the p -th powers of $4t$ smooth parameters. Etaleness identifies the completed local ring at the torsor lift with the completed local ring at its tuple in U^t . Under this identification the thick point is the fibre of $F^t : W^t \rightarrow (W')^t$ over the Frobenius image of that tuple.

In a product of N pure tensors, let m'_i denote the new exponent in slot i . Then

$$0 \leq m'_i \leq N(r-1) < pr, \quad \sum_i m'_i \equiv 0 \pmod{r}.$$

Since p is prime to r , choose integers a_i with

$$r \leq a_i \leq 2r-1, \quad pa_i \equiv m'_i \pmod{r}.$$

Their sum is a multiple of r and is at most $2tr - t < q$. Increase a_1 by the nonnegative multiple $q - \sum_i a_i$ of r . We now have

$$a_i \geq 0, \quad \sum_i a_i = q, \quad pa_i \geq m'_i. \tag{7.18}$$

Multiply the corresponding slot section of $B + m'_i K_W$ by $\eta_W^{(pa_i - m'_i)/r}$. This gives a section of $B + pa_i K_W$, and division by τ^{pa_i} gives exactly its previous torsor expression. In a local canonical frame, the coefficient of τ is a unit. Its p -th power is a nonzero constant on the thick Frobenius fibre, since the p -th powers of all local parameters vanish there.

It follows that the projection-formula map

$$\bigoplus_{\substack{a_i \geq 0 \\ \sum_i a_i = q}} \left(\bigotimes_{i=1}^t H^0(W, \mathcal{O}_W(B + pa_i K_W)) \right) \otimes_{\mathcal{O}_{(W')^t}} \left(- \sum_{i=1}^t a_i \text{pr}_i^* K_{W'} \right) \longrightarrow \mathcal{E}^{\boxtimes t} \tag{7.19}$$

has rank R at one point, hence at the generic point. The products just constructed give the spanning values, up to the nonzero constant frame factors. By finite pushforward and base change these values are the fibre of the target. The identity used in the projection formula is $F^* \mathcal{O}_{W'}(K_{W'}) \simeq \mathcal{O}_W(pK_W)$; it is an identity of line bundles, not the pullback map on differential forms. In compatible local frames its transitions are p -th powers.

The loss of rank on the diagonal. On the diagonal $W' \subset (W')^t$, all the source line bundles in (7.19) equal $\mathcal{O}_{W'}(-qK_{W'})$. Let

$$\Delta_F = \underbrace{W \times_{W'} \cdots \times_{W'} W}_{t \text{ factors}}, \quad h : \Delta_F \rightarrow W', \quad g : \Delta_F \rightarrow W$$

be the common structure map and the first projection, respectively. Both are finite. The line bundle

$$\mathcal{L} = \mathcal{O}_{\Delta_F} \left(\sum_{i=1}^t \text{pr}_i^* B + pq \text{pr}_1^* K_W \right) \quad (7.20)$$

satisfies

$$h_* \mathcal{L} \simeq (\mathcal{E}^{\boxtimes t})|_{W'} \otimes \mathcal{O}_{W'}(qK_{W'}).$$

This is finite base change from F^t , followed by the projection formula; all the line bundles $\text{pr}_i^* \mathcal{O}_W(pK_W)$ are the pullback of $\mathcal{O}_{W'}(K_{W'})$. The columns of (7.19), after twisting, thus restrict to global sections of $h_* \mathcal{L}$. At every point of the diagonal their rank is at most $h^0(\Delta_F, \mathcal{L})$.

The two-factor diagonal filtration and its truncated symmetric pieces are the canonical filtration of [37, Theorem 3.7] and [25, Definition 3.1, Remark 3.2, and Corollary 3.5]. We use the same local construction in $t - 1$ difference slots, and give the resulting ranks and determinant comparison explicitly. Filter $\mathcal{L}$ by powers of the ideal of the reduced diagonal $W \subset \Delta_F$ and push forward by g . The associated pieces are

$$\begin{aligned} \mathcal{G}_j &= \mathcal{O}_W(tB + pqK_W) \otimes A_j(V), \quad V = (\Omega_W^1)^{\oplus(t-1)}, \\ 0 \leq j \leq u_0 &= 4(t-1)(p-1). \end{aligned} \quad (7.21)$$

For completeness, in etale smooth coordinates the other slots add $4(t-1)$ difference variables with their p -th powers zero and no other relations. This follows from the Cartesian relative Frobenius square for an etale chart. Modulo squares the differences transform by ordinary differentials, giving V ; the associated graded algebra is therefore intrinsically $A_\bullet(V)$. The line bundle contributes its restriction $\mathcal{O}_W(tB + pqK_W)$ to each piece. In particular, with $e_j = \text{rk } A_j(V)$,

$$h^0(\Delta_F, \mathcal{L}) \leq \sum_{j=0}^{u_0} h^0(W, \mathcal{G}_j), \quad \sum_{j=0}^{u_0} e_j = p^{4(t-1)} = s^{t-1}. \quad (7.22)$$

Set $n = 4(t-1)$, $G = \max\{0, 2g(C) - 2\}$, and $c_2 = n(n-1)G$. By the retained inequality (7.13), $\mu_{\min}(V|_C) \geq 0$. Lemma 7.2 gives, uniformly in j and p ,

$$\begin{aligned} \mu_{\max}(\mathcal{G}_j|_C) &\leq (tB + pqK_W + (p-1)(t-1)K_W) \cdot C + c_2 \\ &\leq M_p := 2b_0 p l_1 l_2 l_3 H^4 + c_2. \end{aligned} \quad (7.23)$$

Indeed $BH^3 \leq pbLH^3$, $K_W H^3 \geq 0$, and (7.12) bounds the two contributions by $pb_0 H^4$ each, before multiplication by $l_1 l_2 l_3$.

Put $a = l_1 l_2 l_3 H^4$ and $d_i = l_i H \cdot C = l_i a$. Applying Lemma 7.3 to each $\mathcal{G}_j$ yields

$$\begin{aligned} h^0(W, \mathcal{G}_j) &\leq e_j(M_p + 1) \prod_{i=1}^3 (1 + M_p/d_i) \\ &= e_j p^4 ((2b_0)^4 H^4 + O(p^{-1})). \end{aligned} \quad (7.24)$$

The error is uniform in j , since c_2, a, d_i were fixed before p varied. Explicitly, the leading coefficient is $(2b_0)^4 a^4 / \prod_i d_i = (2b_0)^4 H^4$. Summing (7.24), using (7.22), $H^4 \leq 3$, and the strict first inequality in (7.1), gives, for all sufficiently large p ,

$$h^0(\Delta_F, \mathcal{L}) \leq R/4. \quad (7.25)$$

Thus the rank of (7.19) is at most $R/4$ at every point of the diagonal.

The determinant contradiction. Choose R columns of (7.19) with nonzero determinant, and write $a_i^{(\nu)}$ for the weight of slot i in column ν . Their determinant is a nonzero section of the external product of the line bundles

$$\mathcal{Q}_i = (\det \mathcal{E})^{\otimes R/s} \otimes \mathcal{O}_{W'} \left(\sum_{\nu=1}^R a_i^{(\nu)} K_{W'} \right).$$

Let $\lambda_i = R^{-1} \sum_{\nu=1}^R a_i^{(\nu)}$; then $0 \leq \lambda_i \leq q$. Identifying numerical classes through base-field twist, we have

$$\frac{c_1(\mathcal{Q}_i)}{R} = \frac{B}{p} + \left(\frac{p-1}{2p} + \lambda_i \right) K_W. \quad (7.26)$$

The first-Chern coefficient is the usual Frobenius direct-image coefficient; compare [37, Lemma 4.2]. Here only its numerical divisor class is needed. To check it directly, finite base change identifies $F^* \mathcal{E}$ with the pushforward from $W \times_W W$ of the line bundle pulled back from its second slot. The same truncated-diagonal filtration used in (7.21), now with two slots, has pieces $\mathcal{O}_W(B) \otimes A_j(\Omega_W^1)$. Across all $p^4 = s$ truncated monomials, the total exponent of each of the four variables is $s(p-1)/2$. Taking determinants gives

$$c_1(F^* \mathcal{E}) = sB + \frac{s(p-1)}{2} K_W.$$

Relative Frobenius pullback multiplies divisor classes by p after the base-twist identification. Since the determinant of $\mathcal{E}^{\boxtimes t}$ contributes $(\det \mathcal{E})^{R/s}$ in each slot, this proves (7.26).

Use now the base twist of the blow-up of x and of the test class β . By $B/p \leq bL$ on this test, (7.16), and $\lambda_i \leq q$, the normalized class in (7.26) pairs with the test to at most $D^+ \cdot \beta$. Let s_i be any nonzero section of $\mathcal{Q}_i$, and let $m_i = \text{ord}_{x'}(s_i)$. Its divisor has effective strict transform of class $\pi_{x'}^* c_1(\mathcal{Q}_i) - m_i J'$ on the twisted blow-up. The nonnegativity of the test on effective divisors and (7.15) imply

$$m_i(J \cdot \beta) \leq c_1(\mathcal{Q}_i) \cdot \beta \leq RD^+ \cdot \beta < Rd_0(J \cdot \beta).$$

Here the pairings have been identified with their untwisted numerical values. As $J \cdot \beta > 0$, every such section has order strictly less than Rd_0 at x' .

The determinant section is nonzero, so the Künneth formula ensures that each slot section space is nonzero. By Lemma 7.4, its order at $(x', \dots, x')$ is therefore strictly less than

$$Rtd_0 = Rb_0(C_1 + 1) < 3R/4. \quad (7.27)$$

On the other hand, (7.25) bounds the rank of its column matrix at this point by $R/4$. Trivialize the source and target bundles locally, and perform constant invertible row operations so that at least $R - \lfloor R/4 \rfloor$ rows vanish at the point. Every entry of these rows lies in the maximal ideal. Each term of the determinant is consequently in its $(R - \lfloor R/4 \rfloor)$ -th power. The determinant has order at least $3R/4$, contradicting (7.27).

This excludes the assumed Seshadri inequality and proves (7.2). All varieties, line bundles, flags, points, and testing data were fixed before the characteristic grew. Their dependence on X, r, t therefore affects only how large the characteristic must be, whereas the constant $5/b_0$ in the conclusion depends only on C_1 . $\square$

8 Many factors and birational transports

We next turn the quadratic cost bound into a birational obstruction. The argument uses many factors, but their number is fixed before the order of the covering group tends to infinity. This order of choices will be important in the application. Rational evaluation on diagonal products also appears in algebraic formulations of Lie's superposition theorem; compare [5, §§3,6–7 and Theorem 7.1]. We prove the rational evaluation needed here directly from a generically finite forgetting map of degree one.

Proposition 8.1. *There is no sequence of finite cyclic covers*

$$\pi_\nu : Y_\nu \longrightarrow X_\nu$$

of normal projective fourfolds, generically torsors for cyclic groups G_ν of orders $r_\nu \rightarrow \infty$, with the following properties. The group $\text{Bir}(X_\nu)$ is countable, and there are ample rational Cartier divisors L_ν on X_ν and constants $c, C > 0$, independent of ν , such that at very general points

$$\varepsilon(L_\nu) \geq c, \quad \varepsilon(\pi_\nu^* L_\nu) \geq cr_\nu^{1/4}.$$

Moreover, for each fixed integer $t \geq 2$, put

$$Z_{\nu,t} = Y_\nu^t / G_{\nu,\text{diag}}, \quad \theta_{\nu,t} : Z_{\nu,t} \longrightarrow X_\nu^t, \quad P_{\nu,t} = \theta_{\nu,t}^* L_\nu^{\boxplus t}.$$

Here $L^{\boxplus t} = \sum_{i=1}^t \text{pr}_i^ L$. The further required bound is*

$$\varepsilon(P_{\nu,t}) \leq Ct^2$$

at a very general point, for all sufficiently large ν , with this threshold allowed to depend on t .

Proof. Choose an integer T , to be made sufficiently large in terms of c, C . Passing to a tail of the sequence, we may use all the upper bounds with $2 \leq t \leq T$ simultaneously. We suppress ν from the notation until it is necessary to vary the cover. Write $L_Y = \pi^* L$.

Since G is abelian, the finite map θ_t is the quotient by G^t / G_{diag} . Indeed the quotient by the full product group has function field $\mathbb{C}(X^t)$, by the generic torsor property, and the resulting finite birational map to the normal variety X^t is an isomorphism. In particular, P_t is ample.

Curve families and compatible leaves. Set

$$B = CT^2 + 1.$$

On each Z_t , choose a marked covering family of integral curves whose degree divided by multiplicity at the mark is at most B . Such a family is obtained by choosing a curve at a geometric generic

point witnessing a strict bound below $Ct^2 + 1$, and spreading the curve and its mark. We use the geometric generic conventions of Lemma 5.1: finite data can be defined over a countable field, and the relevant generic marked families can be realized over $\mathbb{C}$.

Enlarge these lists by all deck translates, all permutations of the slots, and all nonconstant image families under ordered projections $Z_t \rightarrow Z_s$, for $2 \leq s < t \leq T$. Close under these operations. The lists remain finite, although their sizes need not be bounded uniformly in the covering order.

Projection does not increase the marked curve cost. To see this, let $D \rightarrow D'$ be the nonconstant map of an integral curve to its reduced image, let e be its degree, and let z and z' be the marks. A general local hyperplane through z' , pulled to the normalizations of the curves, has total order $e \text{mult}_{z'} D'$ over z' . The contribution of the branches through z is at least $\text{mult}_z D$. Hence

$$e \text{mult}_{z'} D' \geq \text{mult}_z D.$$

Since P_t minus the pullback of P_s is nef, we obtain

$$\frac{P_s \cdot D'}{\text{mult}_{z'} D'} \leq \frac{P_t \cdot D}{\text{mult}_z D}.$$

After shrinking parameter spaces we may use reduced integral geometric generic image curves. Their projected mark maps remain dominant.

Let H_t be a geometric generic leaf furnished by Lemma 5.1, and put $h_t = \dim H_t$. Lemma 5.4 gives

$$h_t > 0, \quad P_t^{h_t} \cdot H_t \leq (h_t B)^{h_t}. \quad (8.1)$$

Deck invariance and Corollary 5.2 imply that $W_t = \theta_t(H_t)$ is the geometric generic fibre closure of a rational quotient of X^t . We write

$$\zeta_t : X^t \dashrightarrow S_t$$

for such a quotient, with geometrically integral generic fibre. The leaves and their quotients are equivariant under permutations.

The leaves are also compatible with coordinate projections: an upper leaf projects into the lower leaf through the projected tuple. Here is the genericity justification. Retain the parameter space of a curve family when forming its projected image family. Every parameter and endpoint occurring in a generic upper chain is generic in its original parameter space and incidence over the initial field of definition. Its projected endpoint is generic in the image curve. For this lower family, the equality of leaf names at a mark and a generic endpoint is an identity on the generic incidence. Consequently it applies to the projected upper step, regardless of the additional upper history; a stationary projection gives the same equality immediately. A path reaching a generic point of the upper leaf therefore has constant lower name. This proves the asserted inclusion. Whenever different slot sizes are compared below, we take leaves through a generic tuple and its projections and extend their name fields to a common algebraically closed field.

Finite projections to each slot. For r sufficiently large in terms of T, c, C , the map from H_t to each single X -slot is generically finite onto its image. Fix one slot, let its image be I , and write

$$\ell = \dim I, \quad d = h_t - \ell.$$

The image contains an ambient generic point x of X , which is also generic in I over the leaf name. The Seshadri lower bound implies

$$L^\ell \cdot I \geq c^\ell, \quad (8.2)$$

where the assertion is immediate for $\ell = 0$.

Suppose $d > 0$. Over this generic point x , the geometric fibre of $Z_t \rightarrow X$ is Y^{t-1} : choose a lift of x in its torsor to identify the first coordinate. The restriction of P_t is the external sum of the remaining copies of L_Y . Every one of the marked coordinates is generic on Y . Projecting any curve through that mark to a nonconstant slot, with the preceding degree–multiplicity comparison, shows that the Seshadri constant of this restricted polarization is at least $cr^{1/4}$.

Let J be a d -dimensional component of the geometric generic fibre of $H_t \rightarrow I$ containing the chosen generic point. Then

$$P_t^d \cdot J \geq (cr^{1/4})^d. \quad (8.3)$$

We recall the elementary degree bound being used twice here. If an ample rational divisor has Seshadri constant at least a at a smooth point of an e -dimensional subvariety, blow up that point and restrict the nef class consisting of the pulled-back divisor minus a times the exceptional divisor. Its top intersection on the strict transform gives degree at least a^e , since the point has multiplicity one. A limiting argument permits a Seshadri bound given as a supremum. Our image and fibre points are smooth generic points of their subvarieties. The ambient very general bounds continue to hold on the geometric generic data after the field extensions in use.

The pushforward of $P_t^d \cdot H_t$ to I has coefficient equal to the total degree on the geometric generic fibre, at least the contribution of J . Since $P_t - \text{pr}^* L$ is nef, equations (8.2) and (8.3) give

$$P_t^{h_t} \cdot H_t \geq (\text{pr}^* L)^\ell P_t^d \cdot H_t \geq c^\ell (cr^{1/4})^d.$$

For fixed T , this contradicts (8.1) as $r \rightarrow \infty$, because $h_t \leq 4T$. There are only finitely many slot sizes and slots to consider. This proves the assertion simultaneously for all of them on a tail of the sequence. In particular,

$$1 \leq h_t \leq 4. \quad (8.4)$$

A forgetting map of degree one. The same single-slot finiteness holds for W_t . Let D_t denote the degree of its projection to its first-slot image. The degree of H_t over that image contains D_t as a factor, so (8.1) and (8.2) imply

$$D_t \leq \frac{P_t^{h_t} \cdot H_t}{c^{h_t}} \leq KT^8 \quad (8.5)$$

where one may take

$$K = \max_{1 \leq h \leq 4} \left(\frac{h(C+1)}{c} \right)^h.$$

Indeed $h_t \leq 4$, $B \leq (C+1)T^2$, and $T^{2h_t} \leq T^8$. Thus K depends only on c, C , before T is selected.

Projection compatibility and finite first-slot projection give $h_t \leq h_{t-1}$. If these ranks are equal, the image of W_t under forgetting its last slot equals the lower leaf: it is contained in that integral leaf and has the same dimension. The first-slot images therefore agree as well. The degree formula for the two generically finite maps is

$$\frac{D_t}{D_{t-1}} = \deg(W_t \rightarrow W_{t-1}) \in \mathbb{N}. \quad (8.6)$$

All degrees are taken after the common geometric field extension described above, which preserves them.

By (8.4), among the $T-1$ ranks there is a constant stretch of length at least $(T-1)/4$. Choose T , depending only on c, C , so that

$$\frac{T-1}{4} - 1 > \log_2(KT^8).$$

If every adjacent ratio on that stretch were larger than one, (8.6) would force the degrees to grow by a factor at least two at each step, contradicting (8.5). Hence for some $k \geq 3$,

$$h_k = h_{k-1} = h > 0, \quad W_k \longrightarrow W_{k-1} \text{ is birational.} \quad (8.7)$$

Permutation symmetry gives the same conclusion for every omission of one slot, with the corresponding lower leaf. In particular, no finite correspondence of degree greater than one remains in this step.

Rational transport of tangent directions. At a generic tuple of X^k , the tangent space along the leaf W_k is a rational rank- h distribution $\mathcal{D}_k = \ker(d\xi_k)$. It projects injectively to each single-slot tangent space and isomorphically to the lower distribution for every omission. Indeed the corresponding maps of geometric generic leaves are generically finite, and in characteristic zero their differentials have rank h at the generic point. By the equal-rank inclusion already proved, their images are exactly the lower tangent spaces.

Write $A_i \subset T_{X, x_i}$ for the image in slot i . This plane depends rationally only on x_i . To justify the descent, trivialize T_X rationally and use a Grassmann chart. The coordinates of A_i belong to the function field of each $(k-1)$ -tuple containing slot i , because they are obtained from that lower distribution. The intersection of these coordinate subfields is the function field of slot i . Similarly, the graph of the isomorphism from A_i to A_j , obtained by lifting to $\mathcal{D}_k$ and projecting to slot j , depends only on (x_i, x_j) .

For completeness, the coordinate-field intersection can be checked by replacing an omitted coordinate by a fresh independent generic coordinate. A function pulled from the other coordinates is unchanged under that replacement. Repeating for every omitted coordinate, and then specializing the fresh coordinates to general constants, shows that the function belongs to the field of the common coordinates. This applies to the rational chart coordinates of the planes and graphs. It applies also when $k = 3$.

Permutation symmetry thus gives a common rational plane distribution $A(x)$ and isomorphisms

$$R(x, y) : A(x) \xrightarrow{\sim} A(y).$$

Lifting a vector through the same h -plane and using three slots gives the rational cocycle identity

$$R(y, z)R(x, y) = R(x, z). \quad (8.8)$$

Choose a general complex reference point a so that $R(a, -)$ is generically defined and invertible and the specialized cocycle identity holds. Transporting a basis of $A(a)$ gives rational vector fields $v_1, \dots, v_h$ on X , independent at a generic point. Equation (8.8) says exactly that the simultaneous vector fields

$$(v_j, \dots, v_j), \quad 1 \leq j \leq h,$$

span $\mathcal{D}_k$. No commutativity assertion about the v_j is needed.

Rational evaluation and local flows. Equal ranks were enough to construct the tangent transports. We now use the stronger degree-one assertion: it turns evaluation of the missing coordinate into a rational function rather than a finite correspondence. Consider the rational map to its image closure

$$\Lambda = (\text{pr}_{<k}, \xi_k) : X^k \dashrightarrow D.$$

It is birational. Its restriction to a geometric generic leaf of ξ_k is the forgetting map in (8.7), and hence has degree one. More explicitly, let

$$F_0 \subset E \subset F$$

be the function fields of S_k, D, X^k , included by pullback. The field E is generated over F_0 by the first $(k-1)$ -slot functions. Since the geometric generic forgetting map is generically finite, F and E have the same transcendence degree over F_0 ; hence F/E is finite. Geometric integrality of the generic fibre makes F/F_0 regular. Tensoring with an algebraic closure $\bar{F}_0$ gives fields: each tensor is the union of the fields obtained by finite algebraic base extension. Their fraction fields are the function fields of the geometric generic leaf and its projected image. The degree-one geometric projection therefore gives $[F : E] = 1$. This also rules out an extra generic component or an unseen algebraic degree in Λ .

Let ev be the last coordinate of its rational inverse. Choose a general complex tuple (u, y_0) , with $u \in X^{k-1}$, in smooth open sets on which the vector fields, the quotient, and the inverse evaluation are regular. For small $z = (z_1, \dots, z_h) \in \mathbb{C}^h$, let E_z be the ordered composition of the local time- z_j flows of v_j . Let E_z^- be the reversed composition of the negative-time flows. These flows exist holomorphically in both the initial point and the times on sufficiently small neighbourhoods of all chosen coordinates. One may obtain this directly by Picard iteration in smooth coordinate polydiscs: bounded coefficients and derivatives make the integral operator a uniform contraction for small time. Uniqueness gives the opposite-time inverse, and reversing the order gives the inverse of the composition.

The simultaneous flows preserve ξ_k . Put $\lambda(z) = E_z(u)$, acting in all the reference slots. For y near y_0 and sufficiently small z , birational evaluation therefore gives

$$E_z(y) = \text{ev}(\lambda(z), \xi_k(u, y)). \quad (8.9)$$

For each fixed such z , the right side is rational in y . The equality on an analytic open shows that its input lies in D and meets the domain of evaluation; an analytic open is Zariski dense. Thus it defines a rational self-map of X , regular at y_0 after shrinking the neighbourhoods and times.

The same argument gives

$$E_z^-(y) = \text{ev}(E_z^-(u), \xi_k(u, y))$$

on a neighbourhood of y_0 . The resulting rational maps are mutually inverse because they are inverse on a smaller analytic neighbourhood. They are therefore birational self-maps of X . Finally, $z \mapsto E_z(y_0)$ has nonzero derivative, since the fields are independent and $h > 0$. Its image is uncountable, so the birational maps, all regular at y_0 , are uncountably many distinct maps. This contradicts the assumed countability of $\text{Bir}(X)$. $\square$

Lemma 8.2. *Let X be a normal projective terminal variety with $K_X \sim_{\mathbb{Q}} 0$. If a smooth projective resolution of X has irregularity zero, then $\text{Bir}(X)$ is countable.*

Proof. Let $p : W \rightarrow X$ be a smooth projective resolution. Since $H^1(W, \mathcal{O}_W) = 0$, the exponential sequence embeds the analytic Picard group in $H^2(W, \mathbb{Z})$. By GAGA the analytic and algebraic Picard groups agree, so $\text{Pic}(W)$ is countable. Divisor pushforward induces a surjection $\text{Pic}(W) \rightarrow \text{Cl}(X)$: the strict transform of every Weil divisor on X is Cartier on W . Thus $\text{Cl}(X)$ is countable.

Every birational self-map of X is an isomorphism in codimension one. To see this, take a smooth common resolution of its graph, with projections p_1, p_2 to X . Choose a nonzero trivialization η of a sufficiently divisible pluricanonical bundle of X . Its two pullbacks are nonzero regular pluricanonical forms on the common resolution. That section space has dimension one: canonicity gives pullback regularity, and restriction to a big open of X , followed by divisorial extension, identifies the sections with those of the trivial pluricanonical bundle on X . Consequently the two pullbacks are proportional. Terminality says that their zero divisors have support exactly the exceptional prime divisors of p_1 and p_2 , respectively. These supports must coincide. A prime

divisor contracted by the self-map or its inverse would belong to only one of the two supports, which is impossible.

Fix a very ample divisor A on X . Group the birational self-maps according to the Weil linear equivalence class of the strict transform of A back to the source. Within one class, the complete spaces of sections are identified by the codimension-one isomorphisms. The two maps to the embedding of X by $|A|$ therefore differ by a projective linear automorphism preserving that embedded variety. It is enough to prove that this stabilizer has finite effective image on X .

Let S be the identity component of the stabilizer in the projective linear group of the embedding. Its action on X induces a rational action on the smooth projective resolution W . The effective image is a connected linear algebraic group: the kernel of the action is a closed normal subgroup, and a quotient of a linear algebraic group is linear. As proved above, W has a nonzero positive pluricanonical section. Matsumura's obstruction therefore says that this effective image is trivial [30, Corollary 1]. More precisely, that corollary excludes positive-dimensional linear algebraic subgroups of $\text{Bir}(W)$ for a smooth complete variety with a nonzero pluricanonical system. It applies here to the rational action, without requiring that every birational self-map of W be regular.

The identity component of the stabilizer consequently acts trivially. An algebraic group has only finitely many components, so the effective stabilizer is finite. There are countably many possible divisor classes and finitely many birational maps in each such class. Hence $\text{Bir}(X)$ is countable. $\square$

9 Completion of the canonical-index proof

Proof of Theorem 1.1. Apply the canonical cover and the decomposition (4.2) to X . Proposition 4.1 settles all decompositions with at least two positive dimensional blocks and the purely abelian case. The remaining case is a single four-dimensional nonflat factor. Corollary 4.4 settles its noncanonical quotients. It remains to bound the orders when X is canonical and has such a single-factor cover.

Suppose that these orders are unbounded, and choose a sequence whose orders r tend to infinity. Replace each member by its crepant $\mathbb{Q}$ -factorial terminalization V , using Lemma 4.5. The order r is unchanged. The conclusions of Lemma 4.2 are birational properties, so every proper positive dimensional rational image of V has rationally connected resolution, and a smooth projective model of V has irregularity zero.

By Lemma 4.6, any member carrying an effective nonbig divisor of positive Iitaka dimension has canonical order dividing the fixed integer N_{mov} . After discarding a finite initial part of the sequence, every effective Weil divisor of positive Iitaka dimension is therefore big. For the cyclic index cover $\pi : Y \rightarrow V$, Lemma 4.7 now verifies the precise intermediate-fibration hypothesis of Theorem 6.1, both for V with the trivial group and for Y with its cyclic deck group.

Choose an ample rational Cartier divisor L on each V , scaling it by a positive rational number so that $1 \leq L^4 \leq 2$. The scaling is possible without any uniform Cartier denominator. The scalar estimates give constants $C, c > 0$, independent of the member of the sequence, such that at very general points

$$\gamma(L; V) \leq C2^{1/4}, \quad \varepsilon(L; x) \geq c, \quad \varepsilon(\pi^*L; y) \geq cr^{1/4}.$$

For the last inequality we used $(\pi^*L)^4 = rL^4$; the polarization upstairs is ample and invariant under the deck group. Choose once and for all a rational number $C_1 \geq C2^{1/4}$ and put $b_0 = 1/(4(C_1 + 1))$. Then $b_0(C_1 + 1) = 1/4 < 3/4$ and $3(2b_0)^4 < 3/16 < 1/4$, so both strict conditions in (7.1) hold. Proposition 7.1 therefore gives, for every fixed integer $t \geq 2$,

$$\varepsilon(P_t) \leq 20(C_1 + 1)t^2.$$

The constants are independent of the member of the sequence. For each fixed product, its auxiliary choices are made before the characteristic tends to infinity in that proposition. We then use only finitely many such product sizes before letting the canonical orders r tend to infinity in the transport argument.

Lemma 8.2 applies to the terminal V , since $K_V \sim_{\mathbb{Q}} 0$ and its smooth projective model has irregularity zero. Thus $\text{Bir}(V)$ is countable. All hypotheses of Proposition 8.1 have now been checked: the cover is a finite cyclic torsor on a big smooth open, the polarizations have the two displayed lower bounds, the diagonal quotients have the uniform quadratic upper bounds, and the downstairs birational group is countable. That proposition rules out $r \rightarrow \infty$, a contradiction.

The orders are consequently bounded in this final case as well. There are only the cases just treated. Taking a common multiple of their bounded orders gives a single positive integer N_4 with $N_4 K_X \sim 0$ for every input X , as required. $\square$

10 Proof of the uniform Iitaka theorem

Proof of Theorem 1.2. First we deduce nonvanishing from pseudo-effectivity. The companion good-model theorem, with the $\mathbb{Q}$ -factorial dlt choice in its construction [34, Theorem 11.1 and Proposition 2.5], gives a log minimal model (Y, Δ_Y) of (X, Δ) . Its boundary is the strict transform of Δ together with any extracted prime divisors of coefficient one. Hence it is rational, and $D_Y = K_Y + \Delta_Y$ is nef and $\mathbb{Q}$ -Cartier. The companion's rational abundance theorem [34, Theorem 1.1] makes D_Y semiample. On a common smooth resolution

$$X \xleftarrow{p} W \xrightarrow{q} Y$$

with compatible canonical divisors, its comparison lemma [34, Lemma 2.2] gives

$$p^*D = q^*D_Y + E, \quad E \geq 0, \quad E \text{ is } q\text{-exceptional.}$$

Choose a positive integer r such that rD_Y is Cartier and globally generated, and a nonzero section $s \in H^0(Y, \mathcal{O}_Y(rD_Y))$, viewed in the common rational function field. Pulling the effective Cartier divisor $\text{Div}_Y(s) + rD_Y$ back by q , adding rE , and pushing forward by p gives

$$\text{Div}_X(s) + rD \geq 0.$$

The pole test (2.2) therefore gives $0 \neq s \in V_r(D)$. Choose a positive integer t such that trD is integral Cartier. Then s^t is a nonzero section of $\mathcal{O}_X(trD)$, so $\kappa(X, D) \geq 0$. These auxiliary degrees need not be uniform: they only establish the hypothesis for the uniform argument, whose all-degree comparison will retain the full Iitaka field.

Enlarge the fixed coefficient set to $\Phi' = \Phi \cup \{1\}$. Proposition 2.2 gives a semiample dlt model (X', Δ') with coefficients in Φ' and identical rounded section spaces in every integer degree. Write its contraction as $f : X' \rightarrow Z$ and its adjoint as

$$K_{X'} + \Delta' \sim_{\mathbb{Q}} f^*A,$$

where A is ample rational Cartier and $\dim Z = \kappa(X, K_X + \Delta)$.

If $\dim Z > 0$, apply Proposition 3.2 to $f : (X', \Delta') \rightarrow Z$, with total-space dimension four and fixed coefficient set Φ' . Since A is ample, it is big. The proposition gives one degree depending only on Φ' whose every positive multiple is nonempty and generates exactly $\mathbb{C}(Z) = K(D)$ inside

the common rational function field. This same degree covers all positive base dimensions. When $\dim Z = 4$, the contraction is birational and its field is $\mathbb{C}(X')$.

Suppose instead that Z is a point. The adjoint of (X', Δ') is then $\mathbb{Q}$ -linearly trivial. There are three exhaustive cases. If the pair is not klt, the four-dimensional non-klt index theorem [22, Corollary 1.7] gives a common linearly trivial multiple depending only on Φ' . If it is klt and $\Delta' \neq 0$, use [41, Theorem 1.14]. Its hyperstandard coefficient convention includes any prescribed finite rational set: one may take the defining set $\{1 - b : b \in \Phi'\}$ and denominator one. If it is klt and $\Delta' = 0$, use Theorem 1.1. A bound for the positive torsion order gives a common trivial multiple by taking the least common multiple of the finitely many possible orders.

Take a common multiple of these zero-dimensional degrees and of the degree obtained for positive-dimensional bases. The choice depends only on Φ and dimension four. In dimension zero it yields a nonempty system with image a point. In positive dimension, the every-multiple clause of Proposition 3.2 retains the whole Iitaka field. Finally (2.1) transfers the complete systems back to the original pair. This proves every assertion, including the rounded reflexive convention and recovery of the full Iitaka base field. If a canonical representative is replaced by $K_X + \text{Div}(h)$, then the floor divisor changes by $m \text{Div}(h)$ in every integer degree. Multiplication by h^{-m} identifies the two section spaces and leaves all ratios unchanged. The same integer therefore works for all canonical representatives. $\square$

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